

1. O/N/23/22/Q9

In space, an interstellar cloud of dust and gas collapses to form a protostar. The cloud contains hydrogen.

- (a) Describe the energy transfers that take place as the cloud collapses and forms the protostar.

.....
.....
..... [2]

- (b) The collapse results in a nuclear reaction in the protostar that involves the isotope hydrogen-3 (${}^3_1\text{H}$).

The isotope hydrogen-3 (${}^3_1\text{H}$) is radioactive. It decays by beta particle emission.

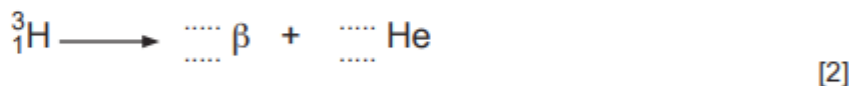
- (i) State how the nuclei of isotopes of the same element are different.

.....
..... [1]

- (ii) Explain why hydrogen-3 cannot decay by alpha particle emission.

.....
..... [1]

- (iii) Complete the equation for the decay of hydrogen-3 to an isotope of helium (He).



- (iv) Explain how the nuclear reaction in the protostar stops further collapse as the protostar becomes a stable star.

.....
.....
.....
.....
..... [3]

- (c) The cloud of dust and gas that collapses also contains atoms of the heaviest elements.

State which part of the life cycle of a star is responsible for the production of the heaviest elements.

.....
..... [1]

2. O/N/23/22/Q10

The isotope iodine-131 emits two types of nuclear radiation. Gamma radiation is one of the types emitted.

A sample of iodine-131 produces a narrow beam of radiation which passes into a magnetic field.

Fig. 10.1 shows some of the paths taken in the magnetic field by the radiation emitted from the sample.

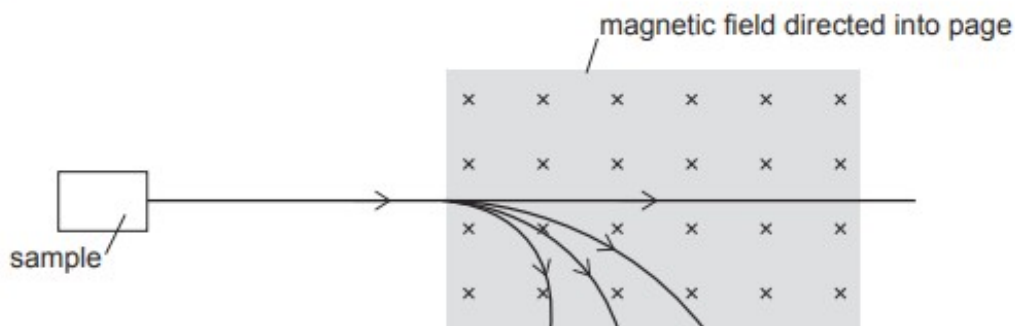


Fig. 10.1

(a) The direction of the magnetic field is into the page.

(i) On Fig. 10.1, mark with a gamma symbol (γ) any of the paths taken by the gamma radiation. [1]

(ii) Determine the other type of radiation emitted by iodine-131.

Explain how you obtain your answer.

type of radiation

explanation

.....
.....
.....

[4]

(iii) In another experiment, the beam of radiation from the sample is incident on a 5 mm thick sheet of lead.

State what happens to the two types of radiation in the beam.

.....
.....
..... [2]

(b) The gamma radiation produced has a frequency of $8.8 \times 10^{19} \text{ Hz}$.

The speed of electromagnetic radiation in a vacuum is $3.0 \times 10^8 \text{ m/s}$.

Calculate the wavelength of this gamma radiation in a vacuum.

wavelength = m [2]

[Total: 9]

3. O/N/23/21/Q8

Uranium-235 ($^{235}_{92}\text{U}$) is the isotope of uranium that is used as the fuel in a nuclear power station.

- (a) The isotope of uranium that is most abundant in nature is uranium-238 ($^{238}_{92}\text{U}$).

State how a neutral atom of uranium-235 differs from a neutral atom of uranium-238.

.....
..... [1]

- (b) In the reactor of a nuclear power station, a nucleus of uranium-235 absorbs a slow-moving neutron and undergoes fission.

- (i) State what happens to the nucleus of an atom during fission.

.....
..... [1]

- (ii) Explain how fission can lead to a chain reaction in the reactor.

.....
.....
.....
.....
.....
.....
.....
..... [4]

- (iii) State the purpose of the moderator in the reactor.

.....
..... [1]

- (iv) Explain the action of the control rods in the reactor.

.....
.....
..... [2]

[Total: 9]

4. M/J/23/22/Q9

A radioactive source emits α -particles, β -particles and γ -radiation.

- (a) (i) State which type of radiation produces the strongest ionising effect.

..... [1]

- (ii) State which type of radiation is deflected most by a magnetic field.

..... [1]

- (b) Fig. 9.1 shows a Geiger-Müller (G.M.) tube and counter. A radioactive source is placed 10 cm from the G.M. tube.

In Fig. 9.2 a piece of metal 5 mm thick is placed between the source and the G.M. tube. The readings on the counter have been corrected for background radiation and show the count rate due to the source.

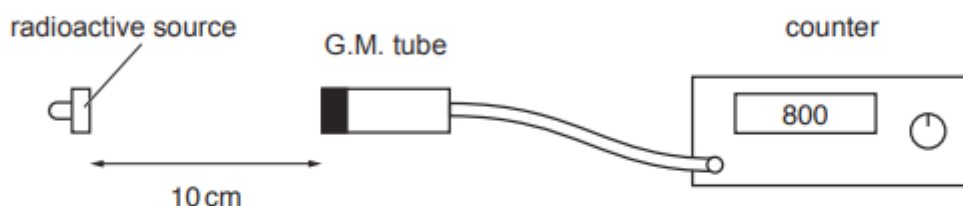


Fig. 9.1

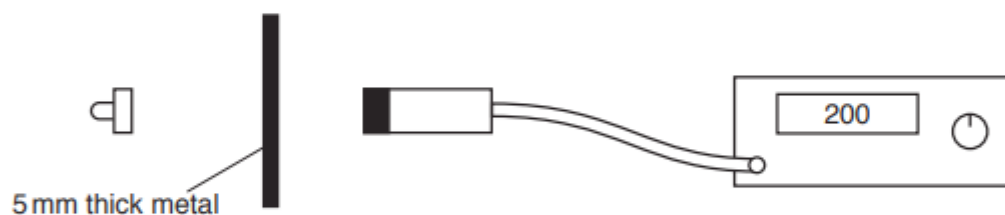


Fig. 9.2

- (i) Explain how the readings show that the source emits β -particles and γ -radiation.

.....
.....
.....
..... [3]

- (ii) State why the readings **cannot** be used to show that the source emits α -particles.

.....
..... [1]

- (c) Describe **one** way that a radioactive source is moved safely in a school laboratory.

.....
..... [1]

5. M/J/23/21/Q10

Carbon-14 is an isotope of carbon. It undergoes radioactive decay with a half-life of 6000 years. The age of a sample of wood is found using the carbon-14 that it contains.

(a) Describe what happens during radioactive decay.

.....
.....
..... [2]

(b) The count rate of the carbon-14 in the sample of wood is initially 1600 counts/minute.

On Fig. 10.1 draw a graph to show how the count rate will vary over the next 24 000 years.

The initial count rate is already marked with an x.

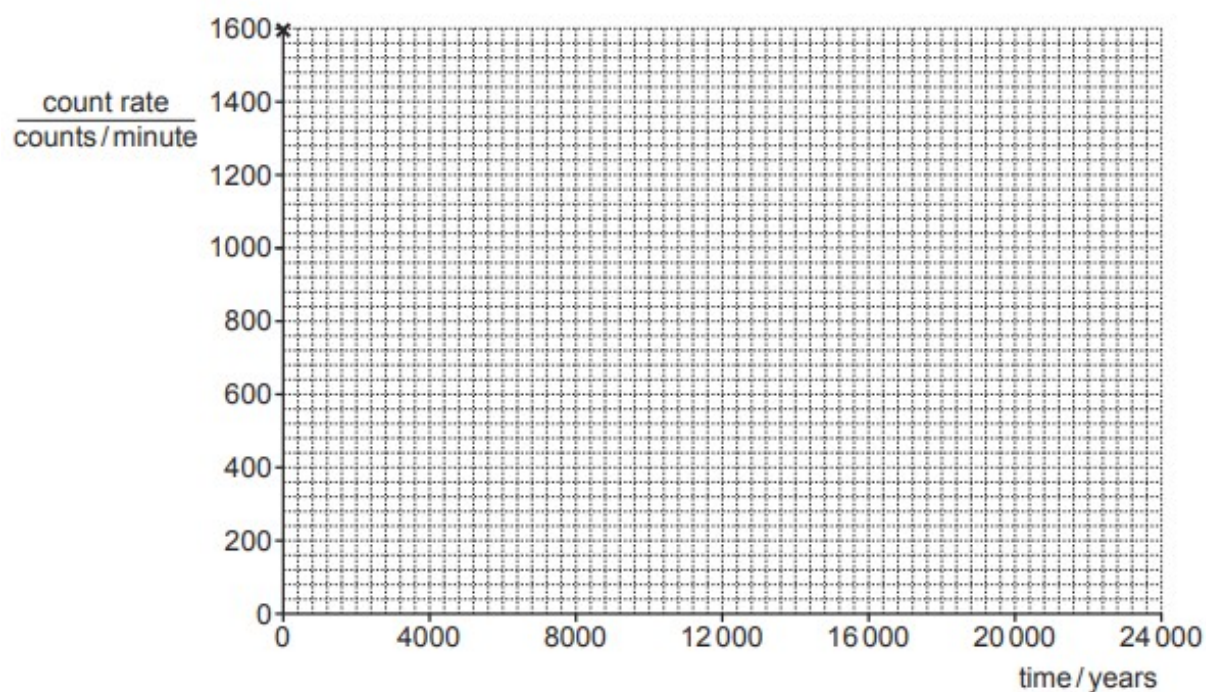


Fig. 10.1

[2]

- (c) Name the apparatus used to measure the radiation from the sample of wood and describe how it is used to measure the count rate.

apparatus used

how the apparatus is used

.....

..... [2]

- (d) Describe how the count rate from a sample of wood is used to find its age.

.....

.....

.....

.....

.....

..... [2]

[Total: 8]

6. O/N/22/22/Q9

Protactinium-234 (${}^{234}_{91}\text{Pa}$) is a radioactive isotope of protactinium that decays to uranium-234 (${}^{234}_{92}\text{U}$).

- (a) Compare the nuclide notation ${}^{234}_{91}\text{Pa}$ with the nuclide notation ${}^{234}_{92}\text{U}$ and deduce what this shows about what is emitted from a nucleus of protactinium-234 as it decays to uranium-234.

Place a tick (✓) in the appropriate boxes of Table 9.1 to show what is deduced from comparing the nuclide notations.

Table 9.1

	yes	no	it is not possible to tell
an alpha-particle is emitted			
a beta-particle is emitted			
a gamma ray is emitted			

[2]

- (b) The most abundant isotope of protactinium is protactinium-231.

- (i) Explain, by referring to their nuclear compositions, why protactinium-231 and protactinium-234 are both isotopes of the same element.

.....
 [1]

- (ii) Explain, by referring to their nuclear compositions, why protactinium-231 and protactinium-234 are different isotopes of that element.

.....
 [1]

- (c) A teacher places a radiation detector on a bench in a school laboratory and switches it on.

- (i) The teacher measures and records the background radiation count rate.

Describe what is meant by 'background radiation' and state **two** significant sources of the count rate recorded by the teacher.

background radiation

.....

source 1

source 2

[3]

- (ii) The teacher moves a sample of protactinium-234 so that it is next to the detector.

Suggest **one** precaution that ensures that the sample is moved in a safe way.

.....
..... [1]

- (iii) The count rate is measured every 20 s with the sample present, and then corrected for background radiation.

Fig. 9.1 shows a graph of the corrected count rate against time for the protactinium-234 sample.

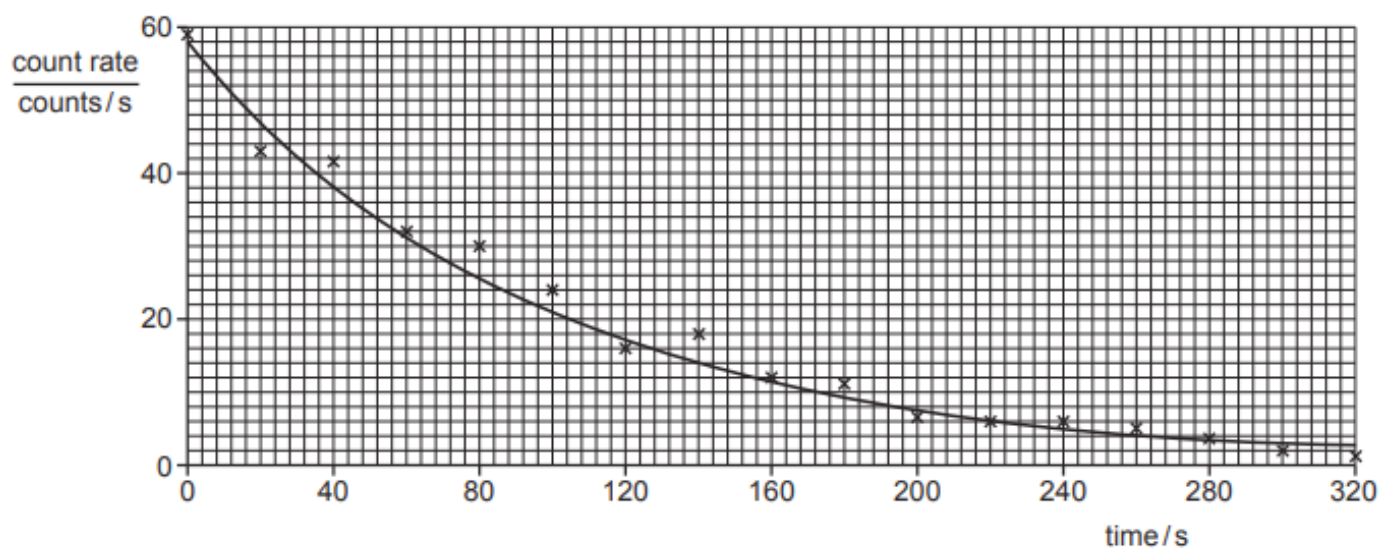


Fig. 9.1

The curve is the best-fit line.

Explain why many of the crosses do not lie on the curve.

.....
.....
..... [2]

- (iv) Using Fig. 9.1, determine the half-life of protactinium-234.

Show your working.

half-life = [3]

- (v) The uranium-234 formed from the protactinium-234 is also radioactive. Its half-life is many thousands of years.

Explain why the radiation from uranium-234 does **not** affect the count rates measured in this experiment.

.....

.....

..... [2]

[Total: 15]

7. M/J/22/22/Q9

Thorium-229 is a radioactive isotope used in several medical applications that involve alpha-particles and beta-particles.

- (a) During ionisation, a helium atom becomes a helium ion.

Fig. 9.1 shows a diagram of a helium (He^+) ion.

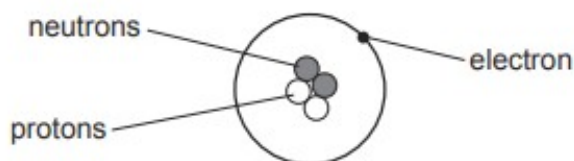


Fig. 9.1

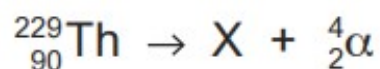
- (i) State how the structure of a helium atom differs from the structure of the helium ion.

.....
 [1]

- (ii) State how the structure of an alpha-particle differs from the structure of the helium ion.

.....
 [1]

- (iii) A nucleus of thorium-229 ($^{229}_{90}\text{Th}$) decays by alpha (α) emission to a nucleus of element X.



The nucleus of X then decays to a nucleus of Y by beta (β) emission.



Complete Table 9.1 to show the number of protons and neutrons in a nucleus of X and in a nucleus of Y.

Table 9.1

nucleus	number of protons	number of neutrons
X		
Y		

[4]

- (b) Experiments can show that a sample of a material is radioactive.

Describe the apparatus and the procedure used to show that a sample emits **both** alpha-particles and beta-particles.

You may draw a diagram of the apparatus, if you wish.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (c) (i) State what is meant by the half-life of thorium-229.

.....

..... [2]

- (ii) A sample of pure thorium-229 contains 4.0×10^{14} atoms. After 22 000 years, the number of atoms of thorium-229 in the sample is 5.0×10^{13} .

Determine the half-life of thorium-229.

Show your working.

half-life = [3]

8. M/J/22/21/Q9

Table 9.1 shows details of seven different nuclides.

Table 9.1

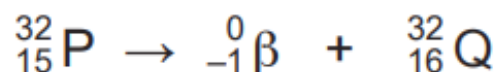
nuclide	radiation emitted	half-life
hydrogen-2	none	–
hydrogen-3	beta	12 year
francium-223	beta	22 min
iridium-192	gamma	74 day
phosphorus-32	beta	14 day
radon-222	alpha	4 day
technetium-99	gamma	6 hour

- (a) (i) Hydrogen-2 and hydrogen-3 are isotopes of the element hydrogen.

Define the term isotope.

.....
 [2]

- (ii) The equation for the decay of phosphorus-32 (P-32) as it emits a beta particle is:



Explain whether the equation shows that Q is another isotope of phosphorus. Q is not the chemical symbol for the atom.

.....
 [1]

- (b) One of the sources in Table 9.1 is used in a medical procedure to detect unusual bone structures. It is injected into a patient and the radiation emitted is detected outside the body.

- (i) State which source in Table 9.1 is most suitable for this type of medical use.

..... [1]

- (ii) Explain **two** reasons for your choice

.....

 [2]

(c) Radon gas is one natural source of background radiation.

Some causes of background radiation are man-made, for example, X-rays.

(i) State one other **natural** source of background radiation.

..... [1]

(ii) State one other **man-made** source of background radiation.

..... [1]

(iii) State one **harmful** effect of background radiation.

..... [1]

(d) One fusion reaction that occurs is:



(i) Complete the equation to show the missing proton and nucleon numbers. [1]

(ii) Deduce the name of particle X.

..... [1]

(iii) Suggest where this fusion reaction takes place.

..... [1]

(e) Compare the properties of alpha-particles and beta-particles in terms of their:

- ability to penetrate through materials
- ionising effects
- deflection in a magnetic field.

.....
.....
.....
.....
.....
.....
..... [3]

[Total: 15]

9. O/N/21/22/Q6

Isotope X is radioactive. It decays by alpha-particle emission to a stable isotope.

- (a)** State how a nucleus of X changes when it emits an alpha-particle.

.....

.....

..... [2]

- (b)** There is a radiation detector in a laboratory where there are no radioactive samples.

The detector is switched on and shows an average count rate of 22 counts/minute.

- (i)** State why the radiation detector shows a count rate.

.....

..... [1]

- (ii)** A sample of isotope X is placed 2cm from the detector and the reading displayed is 8000 counts/minute.

The sample is moved a distance of 10cm from the detector. The reading returns to an average value of 22 counts/minute.

Explain why the reading returns to the original value.

.....

.....

..... [2]

- (c)** An alpha-particle passes into a region where there is a magnetic field. In the magnetic field, a force acts on the alpha-particle so that it follows a circular path. Fig. 6.1 shows that the particle passes through point J.

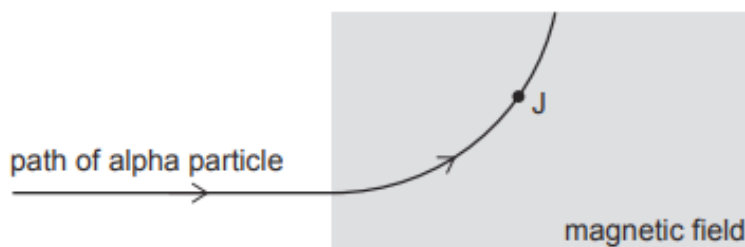


Fig. 6.1

- (i)** On Fig. 6.1, draw an arrow through point J to show the direction of the force on the alpha-particle at J. [1]

- (ii) Determine the direction of the magnetic field and mark a tick in the box (✓) that indicates this direction.

☐	to the left
☐	to the right
☐	towards the top of the page
☐	towards the bottom of the page
☐	into the page
☐	out of the page

[1]

- (iii) Explain whether this force does work on the alpha-particle as the particle moves along the circular path.

.....

.....

..... [2]

[Total: 9]

10. O/N/21/21/Q10

- The isotope yttrium-90 ($^{90}_{39}\text{Y}$) is radioactive. It is a beta-particle emitter that decays to product Q. Product Q is stable.

(a) State one feature that is common to all isotopes of yttrium.

..... [1]

(b) Describe how a **neutral atom** of Q differs from a **neutral atom** of yttrium-90.

.....

 [3]

(c) A sample of yttrium-90 is placed close to a radiation detector in a laboratory. There are no other radioactive samples in the laboratory. A counter records the count rate.

Fig. 10.1 is a graph of the count rate plotted against time.

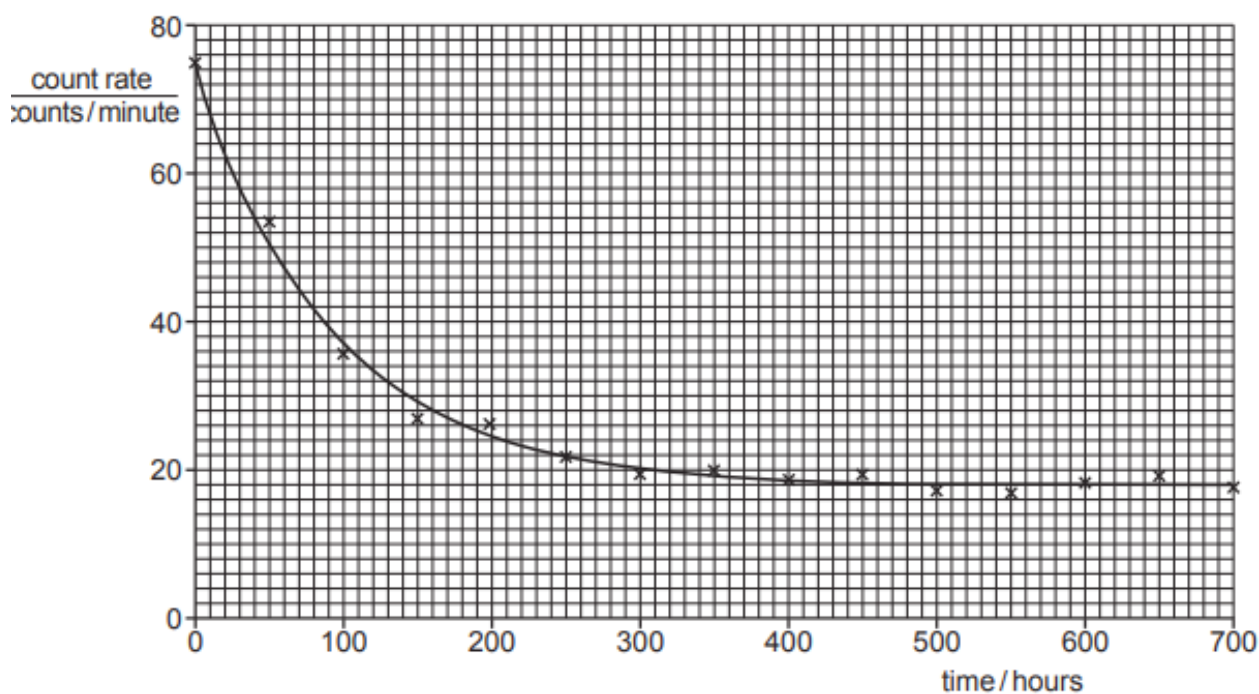


Fig. 10.1

(i) Using Fig. 10.1, determine the average background count rate.

average background count rate = [1]

(ii) Suggest **two** different origins for the background count.

1.

2.

[2]

(iii) Using Fig. 10.1, determine the half-life of yttrium-90. Show how the answer is obtained.

half-life = [4]

(iv) Many of the points plotted in Fig. 10.1 do not lie on the best-fit line.

Explain why.

.....

.....

..... [2]

(d) A beam of beta-particles, travelling in a vacuum, enters the region between two parallel, metal plates. One plate is negatively charged and the other is positively charged.

Fig. 10.2 shows the arrangement.

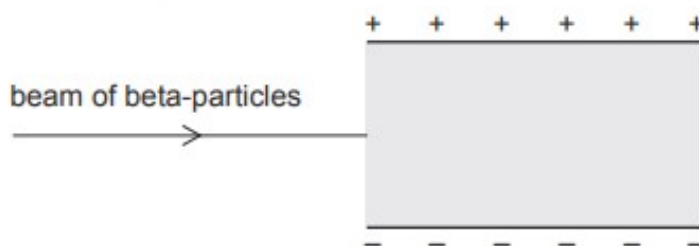


Fig. 10.2

On Fig. 10.2, draw the path taken by the beta-particles as they travel between the two plates.

[2]

11. M/J/21/22/Q11

A highly radioactive source that emits beta-particles is placed a few centimetres away from a detector, as shown in Fig. 11.1.

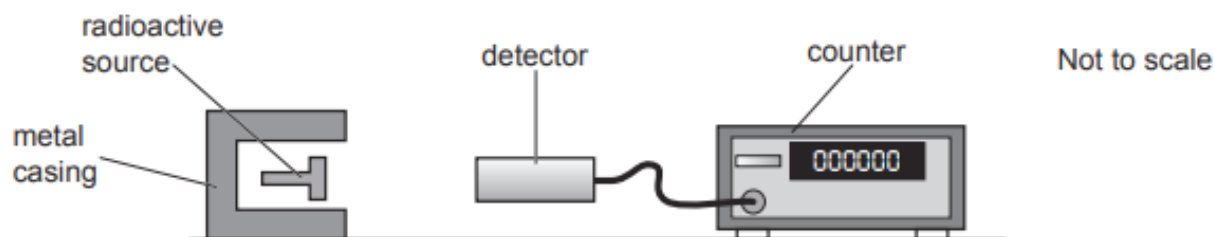


Fig. 11.1

(a) State the name of the particle which has the same mass and charge as a beta-particle.

..... [1]

(b) State and explain why the metal casing in Fig. 11.1 is used.

.....
..... [2]

(c) State and explain what happens to the number of particles detected in a minute as the radioactive source is moved:

(i) a few centimetres **further** away from the source

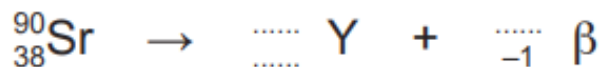
.....
.....
..... [2]

(ii) more than a metre away from the source.

.....
.....
..... [2]

- (d) A nucleus of strontium-90 (Sr-90) decays by beta emission to a nucleus of yttrium (Y).

Complete the decay equation for this decay.



[3]

- (e) Nuclear fusion and nuclear fission both release large amounts of energy.

- (i) Describe how the process of nuclear fusion differs from the process of nuclear fission.

.....

.....

.....

.....

..... [3]

- (ii) Describe the conditions needed for nuclear fusion to take place.

.....

.....

.....

..... [2]

[Total: 15]

12. M/J/21/21/Q10

Table 10.1 contains details of the nature and some properties of alpha, beta and gamma emissions.

Table 10.1

	alpha	beta	gamma
nature	2 protons and 2 neutrons		
charge		negative	
penetrating power		stopped by 5 mm of aluminium	

- (a) Complete Table 10.1 by filling in the missing details. [6]
- (b) Surgical instruments in sealed plastic bags are placed in thin plastic boxes. A conveyor belt takes the boxes close to a cobalt-60 source which sterilises the instruments.

This is shown in Fig. 10.1.

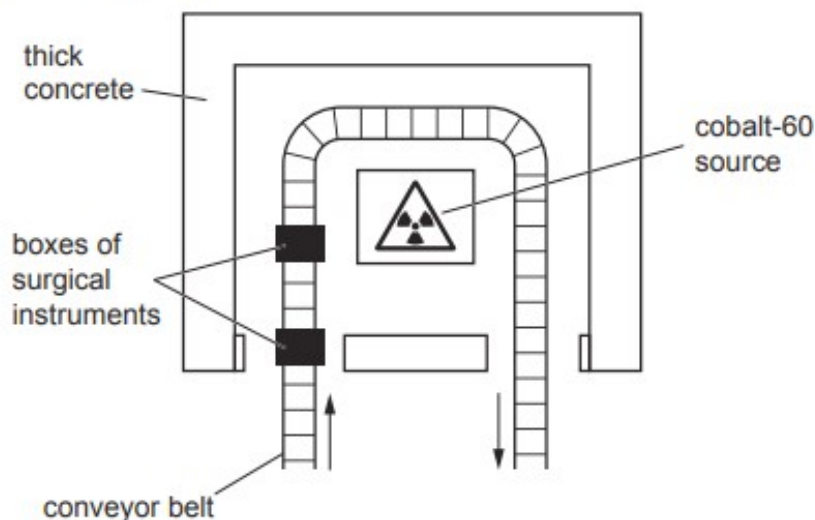


Fig. 10.1 (viewed from above)

The cobalt-60 source is a radioactive isotope of cobalt that emits gamma-radiation.

- (i) Describe what is meant by the term *isotope*.

.....
 [2]

- (ii) Suggest a property of gamma-radiation that enables it to sterilise the instruments in the bags in the boxes.

..... [1]

(iii) State why a source emitting only alpha-radiation cannot be used in this way.

.....
..... [1]

(iv) The half-life of cobalt-60 is 5.3 years.

Explain why a source with a half-life of 5.3 minutes is unsuitable for use in this application.

.....
..... [1]

(c) Geiger and Marsden performed an experiment in which alpha-particles were fired at a thin film of gold.

(i) Fig. 10.2 shows an alpha-particle passing close to the nucleus of a gold atom.

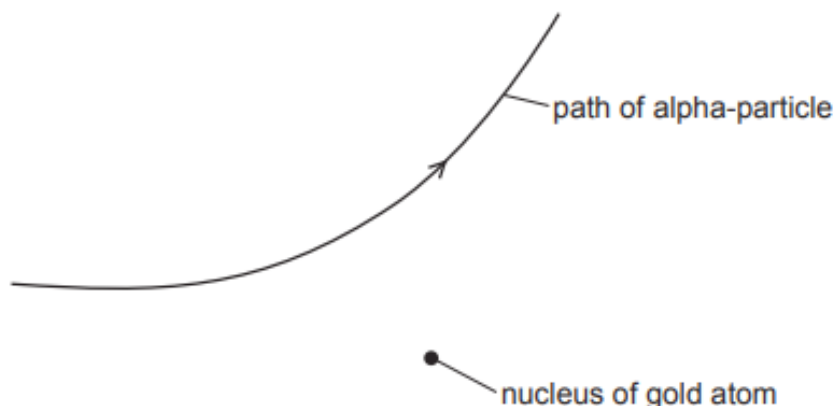


Fig. 10.2

Explain why the alpha-particle is deflected.

.....
..... [2]

(ii) In the experiment, most of the alpha-particles pass straight through the foil without deflection.

Explain, using ideas about the structure of the atom, why this happens.

.....
..... [2]

[Total: 15]

13. O/N/20/22/Q6

Phosphorus-32 ($^{32}_{15}\text{P}$) is an isotope of phosphorus that undergoes radioactive decay.

(a) The most common isotope of phosphorus is phosphorus-31.

(i) Describe the structure and composition of a neutral atom of phosphorus-31.

.....

 [2]

(ii) State how an atom of phosphorus-32 differs from an atom of phosphorus-31.

.....
 [1]

(b) Phosphorus-32 decays by beta-particle emission to a stable isotope of sulfur. The half-life for this decay is 2.0 weeks.

(i) State how a nucleus of this isotope of sulfur is different to a nucleus of phosphorus-32.

.....
 [1]

(ii) At time $t=0$, a radioactive sample contains 3.2×10^{11} atoms of phosphorus-32.

At the same moment, the sample contains no atoms of sulfur. This is shown by the cross on Fig. 6.1.

On Fig. 6.1, plot a graph to show how the total number of sulfur atoms in the sample changes with t and draw a suitable curve. [3]

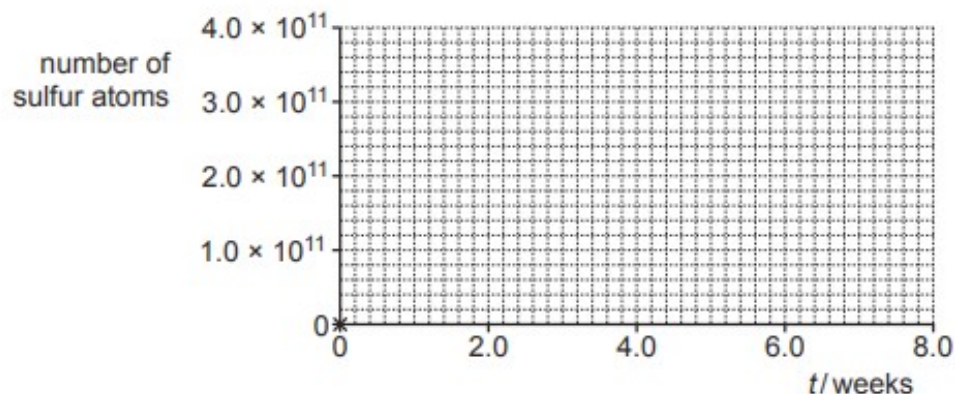


Fig. 6.1

(c) State **two** precautions taken when storing or moving radioactive materials.

1.

.....

2.

.....

[2]

[Total: 9]

14. M/J/20/22/Q7

Radon-222 is an isotope of radon which undergoes a series of radioactive decays.

Fig. 7.1 is a diagram showing the proton number (atomic number) and nucleon number (mass number) of nuclei involved in the series of decays.

The point P represents a nucleus of radon-222.

Starting at P, a nucleus of radon-222 decays to Q; then from Q to R; then from R to S; then from S to T; and finally from T to U.

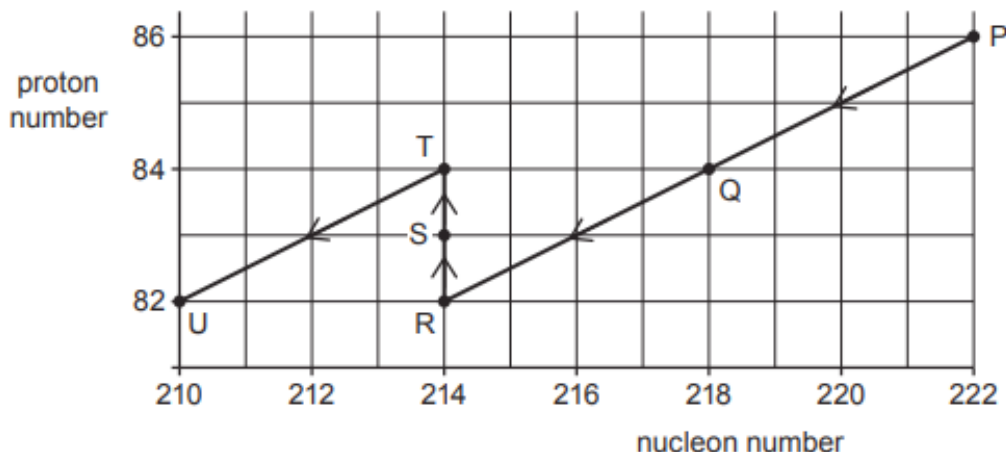


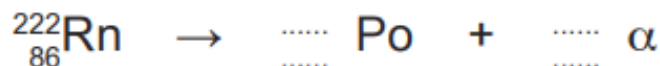
Fig. 7.1

- (a) (i) State two points on Fig. 7.1 which represent isotopes of the same element.
..... [1]
- (ii) Different isotopes of the same element have different atomic compositions.
State how the composition of their atoms is different.
.....
..... [1]
- (b) A nucleus of radon-222 emits an alpha-particle as it decays.
- (i) The radioactive decay of a single nucleus is random.
Explain what is meant by the *random* radioactive decay of a nucleus.
.....
.....
..... [1]

- (ii) In nuclide notation, radon-222 is written as $^{222}_{86}\text{Rn}$.

When a nucleus of radon-222 emits an alpha-particle (α), it decays to an isotope of polonium (Po).

Complete the decay equation below for this decay.



[2]

- (c) (i) State the name of the particle emitted as a nucleus of R decays to a nucleus of S.

..... [1]

- (ii) Describe the change in the composition of a nucleus of R as it decays to a nucleus of S.

.....

.....

..... [2]

[Total: 8]

15. M/J/20/21/Q7

Two isotopes of carbon are carbon-12 and carbon-14.

One of these isotopes, carbon-14, undergoes radioactive decay.

- (a) Describe what is meant by *radioactive decay*.

.....
 [2]

- (b) Carbon-12 has a proton number (atomic number) of 6 and a nucleon number (mass number) of 12.

Complete Table 7.1 for a neutral atom of each of these two isotopes.

Table 7.1

	carbon-12	carbon-14
number of protons	6	
number of neutrons		
number of electrons		

[2]

- (c) A sample of carbon-14 is contained in a thin aluminium container of thickness 0.2 mm.

Radiation from the sample is detected outside the container.

When the thickness of the aluminium is increased to 6 mm, no radiation from the sample is detected outside the container.

- (i) State the type of radiation which is stopped by increasing the thickness of the aluminium.

..... [1]

- (ii) Explain how you know that the sample does **not** emit one other type of radiation.

.....

 [2]

[Total: 7]

16. O/N/19/22/Q10

All the isotopes of the gas radon are radioactive.

- (a) State **one** similarity and **one** difference between the nuclei of two different isotopes of radon.

similarity

difference

[2]

- (b) The isotope radon-222 decays by alpha-particle emission to an isotope of polonium (Po). The proton number (atomic number) of polonium is 84.

- (i) Determine the number of neutrons in an atom of the polonium isotope.

number of neutrons = [2]

- (ii) Determine the number of protons in an atom of radon-222.

number of protons = [2]

- (iii) Describe how a neutral atom of helium (${}^4_2\text{He}$) differs from an alpha-particle.

.....

..... [1]

- (c) In an experiment to collect a small quantity of helium, a sample of radon-222 is enclosed in an inner glass tube which has a very thin wall. Fig. 10.1 shows that this tube is placed inside a container that is initially evacuated.

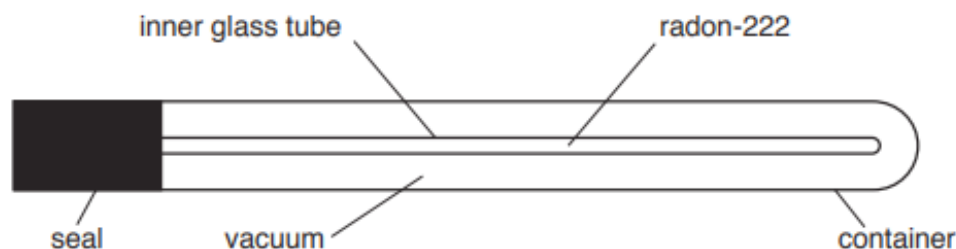


Fig. 10.1

Both the container and the inner glass tube are sealed.

As the radon-222 decays, alpha-particles pass through the thin wall of the inner glass tube.

Fig. 10.2 shows how the total number of alpha-particles produced by the radioactive decay of the radon-222 changes as time passes.

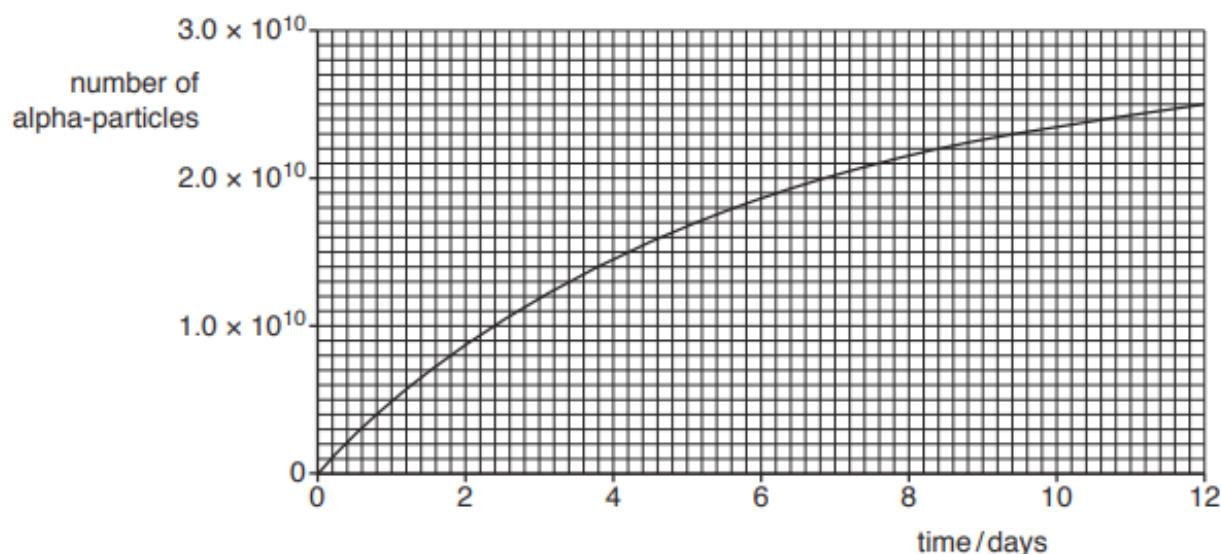


Fig. 10.2

- (i) Use Fig. 10.2 to determine the number of alpha-particles produced in 7.6 days.

..... [1]

- (ii) Initially, there are 2.8×10^{10} atoms of radon-222 in the inner glass tube. As each radon-222 atom decays, it produces an alpha-particle.

Calculate the number of radon-222 atoms that remain after 7.6 days.

number of atoms remaining = [1]

- (iii) Using the number of radon-222 atoms present initially and the number present after 7.6 days, calculate the half-life of radon-222.

half-life = [3]

- (d) The alpha-particles become helium atoms which are collected in the vacuum shown in Fig. 10.1.

Explain, in terms of the properties of alpha-particles, why the wall of the inner glass tube must be extremely thin.

.....

..... [1]

- (e) In some parts of the world, radon-222 accumulates in the air in buildings and is breathed in by people.

Explain why the presence of an alpha-emitter in the lungs is particularly hazardous.

.....

.....

..... [2]

[Total: 15]

17. O/N/19/21/Q7

Table 7.1 shows the composition of four neutral atoms and one atom that has been ionised.

These five particles are labelled P, Q, R, S and T.

Table 7.1

	P	Q	R	S	T
number of electrons	17	18	18	19	20
number of neutrons	20	20	24	23	21
number of protons	17	19	18	19	20

(a) The charge on an electron is $-1.6 \times 10^{-19} \text{C}$.

(i) State which of P, Q, R, S and T is the ionised atom.

..... [1]

(ii) State the size and the sign of the charge on the ionised atom.

charge = [1]

(b) State which pair from P, Q, R, S and T are isotopes of the same element.

..... [1]

(c) One of the atoms is a radioactive isotope that decays to produce one of the other atoms.

(i) Deduce which atom decays in this way.

..... [1]

(ii) Deduce which atom is the product of this decay.

..... [1]

(iii) Deduce which type of radioactive decay occurs.

..... [1]

(iv) The half-life for this radioactive decay is 33 years.

A radioactive sample contains a large number of the atoms that decay in this way.

Determine the fraction of this number of atoms that will decay in 99 years.

fraction that decays = [3]

18. M/J/19/22/Q11

- (a) Stars are formed inside a cloud of gas and dust in space.

When a star forms, the cloud first becomes smaller as it collapses.

- (i) State what causes the gas cloud to collapse.

..... [1]

- (ii) Eventually nuclear fusion occurs in the star.

State **two** conditions required for nuclear fusion to occur in the star.

1.

2.

[2]

- (b) When a large star explodes, nuclei of heavy elements are formed.

One such nucleus is uranium-235 ($^{235}_{92}\text{U}$).

- (i) State the name and the number of each type of particle found in one nucleus of $^{235}_{92}\text{U}$.

name of particle number found in nucleus

name of particle number found in nucleus

[2]

- (ii) State one feature that is the same for the nuclei of all isotopes of uranium.

.....

..... [1]

- (iii) A nucleus of $^{235}_{92}\text{U}$ in a nuclear reactor can undergo nuclear fission.

Describe the process of nuclear fission in a nuclear reactor.

.....

.....

..... [2]

- (iv) State one difference between nuclear fusion and nuclear fission.

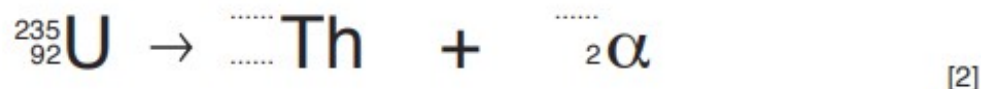
.....

..... [1]

- (v) The nucleus of $^{235}_{92}\text{U}$ is not stable.

It decays by the emission of an alpha-particle (α) into a nucleus of thorium (Th).

Complete the nuclear equation that represents this decay.



- (c) Geiger and Marsden carried out an experiment to investigate the scattering of alpha-particles. Fig. 11.1 shows a diagram of the apparatus that they used.

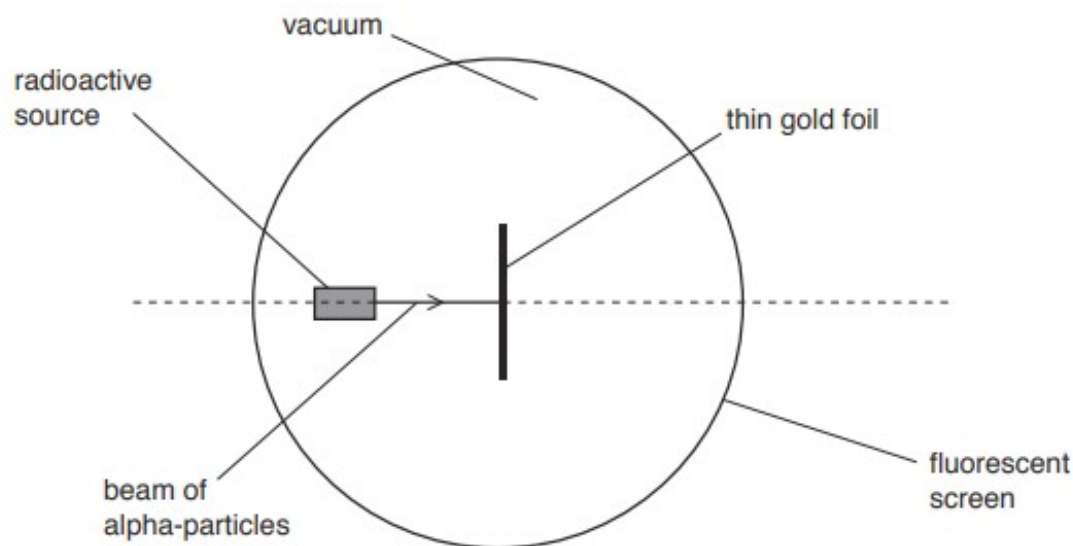


Fig. 11.1

When an alpha-particle strikes the fluorescent screen a flash of light is emitted.

- (i) On Fig. 11.1, mark with a letter P where most flashes occur on the screen. [1]

- (ii) Describe what is seen on the other parts of the screen.

.....
 [1]

- (iii) State two conclusions about atoms of gold that resulted from this experiment.

.....

 [2]

[Total: 15]

19. M/J/19/21/Q11

A teacher carries out an experiment on radioactivity using the apparatus shown in Fig. 11.1.

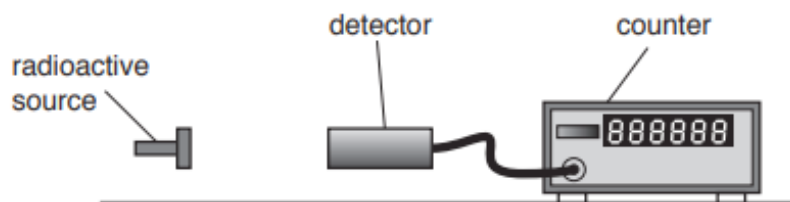


Fig. 11.1 (not to scale)

The count rate is a measurement of the count during one minute.

(a) (i) State the name of one type of radiation detector.

..... [1]

(ii) State one safety precaution the teacher must take when using a radioactive source.

.....
..... [1]

(b) The teacher states that the emissions from the radioactive source are random in time.

(i) State what is meant by *random in time*.

.....
..... [1]

(ii) Describe how the apparatus is used to show that the emissions are random in time.

.....
..... [2]

(c) (i) State what is meant by *background radiation*.

.....
..... [1]

(ii) The teacher measures the count rate due to the source.

Describe how the count rate is corrected for the background radiation.

.....
.....
..... [2]

- (d) The corrected count rate is recorded for various distances d between the source and the detector. Fig. 11.2 shows a graph of the results obtained.

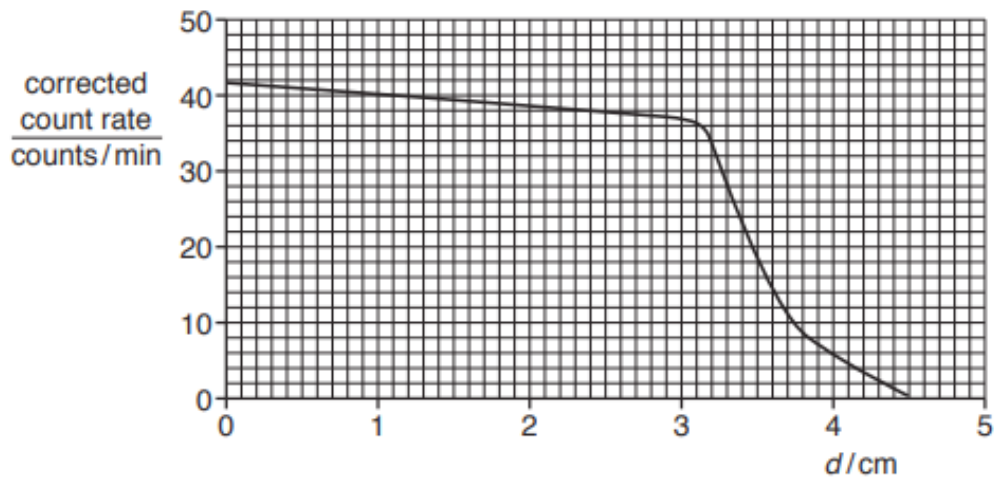


Fig. 11.2

- (i) Determine the maximum distance that the radiation travels in air.

..... [1]

- (ii) State the name of the radiation emitted by the radioactive source and give a reason for your answer.

radiation is

reason

..... [2]

- (iii) As the radiation from the source travels in the air it causes ionisation.

Describe what is meant by *ionisation* of the air.

.....

.....

..... [1]

- (e) The half-life of another radioactive source is 40 minutes.

Initially, the corrected count rate from this source is 300 counts/minute.

On Fig. 11.3, plot suitable points and draw a graph of the corrected count rate as it changes with time.

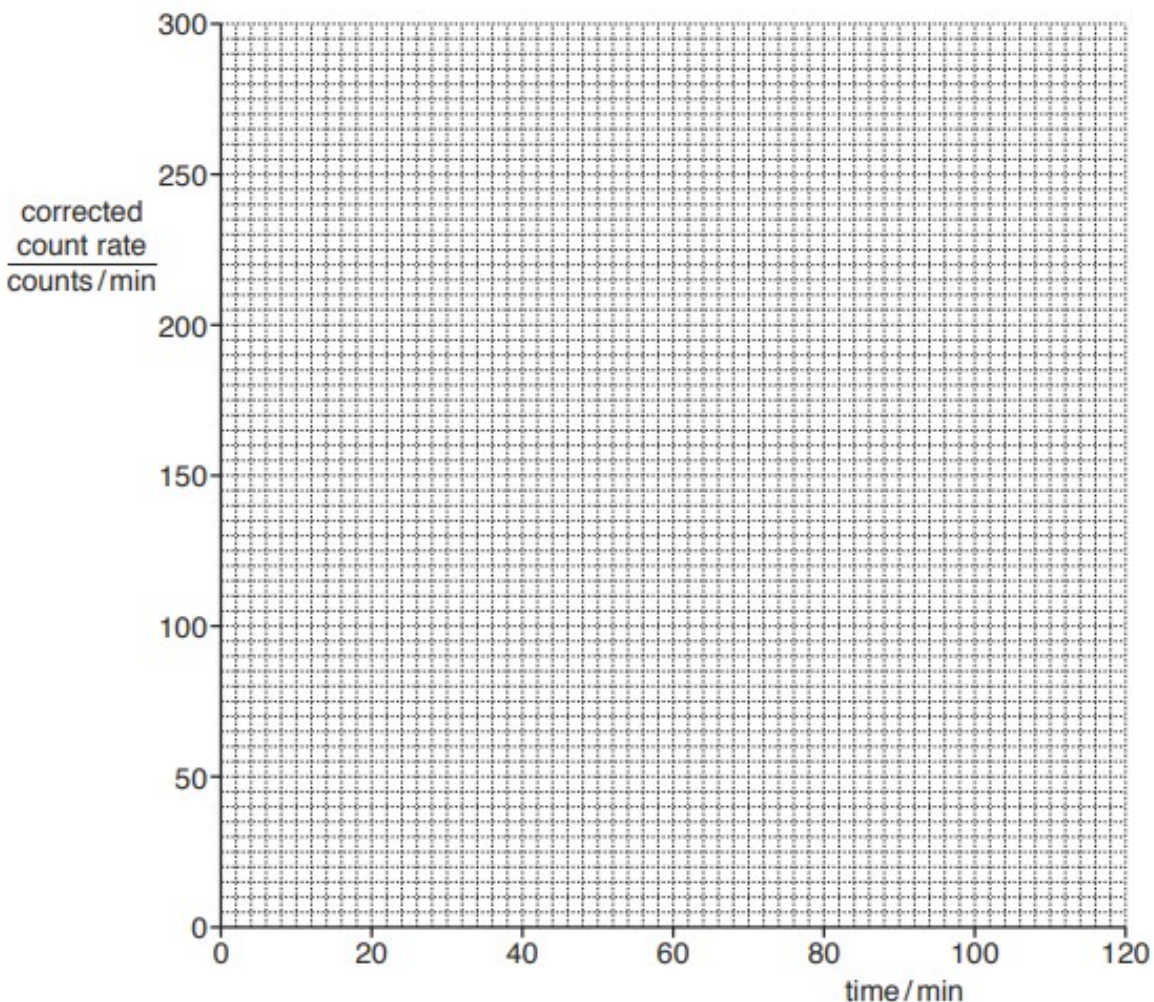


Fig. 11.3

[3]

[Total: 15]

20. O/N/18/22/Q6

Fig. 6.1 shows a radiation detector placed on a laboratory bench.

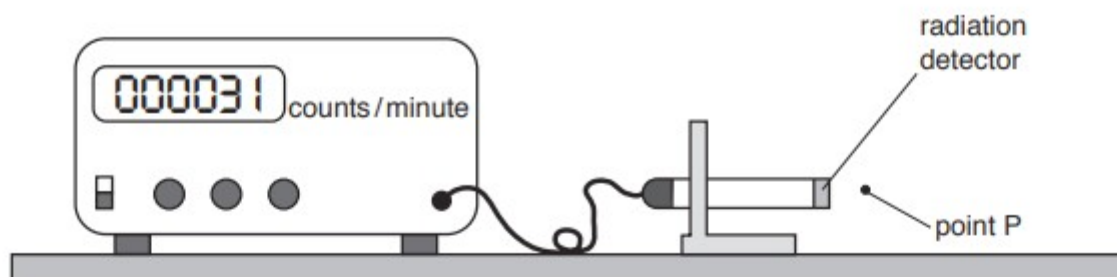


Fig. 6.1

The detector is switched on and six readings of the count rate are recorded.

The table in Fig. 6.2 shows the readings obtained.

reading number	1	2	3	4	5	6
$\frac{\text{count rate}}{\text{counts/minute}}$	31	36	29	32	31	33

Fig. 6.2

- (a) Using all the readings obtained, determine an average value for the background count rate.

background count rate = [1]

- (b) Fig. 6.1 shows a point P which is a very short distance from the end of the radiation detector. A sample of the radioactive isotope cobalt-60 is placed at P. The average value of the count rate obtained is now 975 counts/minute.

The average count rate is determined with different objects between the radiation detector and the sample. The table in Fig. 6.3 shows the results obtained.

object	$\frac{\text{average count rate}}{\text{counts / minute}}$
no object	975
four sheets of paper	976
0.50mm thickness sheet of aluminium	117
2.0cm thickness sheet of lead	52

Fig. 6.3

(i) Indicate, by placing ticks (✓) in the appropriate boxes, the radiation emitted by cobalt-60.

☐

alpha-particles

☐

beta-particles

☐

gamma rays

[1]

(ii) This radiation is produced when a nucleus of cobalt-60 ($^{60}_{27}\text{Co}$) decays into a nucleus of the daughter product X. Product X is not radioactive.

Determine

1. the number of protons in a nucleus of X,

number of protons = [1]

2. the number of neutrons in a nucleus of X.

number of neutrons = [1]

(c) The half-life of cobalt-60 is 5.3 years.

(i) State what is meant by *half-life*.

.....

 [2]

(ii) When there is a lead sheet between the detector and the sample, the average count rate is obtained from six readings taken at one-minute intervals. The six readings are given in the table in Fig. 6.4.

reading number	1	2	3	4	5	6
$\frac{\text{count rate}}{\text{counts/minute}}$	61	46	43	56	49	57

Fig. 6.4

There are reasons for suggesting that the variation in these readings is random and not because the number of cobalt-60 atoms in the sample is decreasing.

State two of these reasons.

1.

 2.
 [2]

21. O/N/18/21/Q10

- (a) Two neutral atoms are atoms of different isotopes of the same element.

State

- (i) one similarity in the structures of these two atoms,

..... [1]

- (ii) one difference between the structures of these two atoms.

..... [1]

- (b) The isotope plutonium-239 decays by the emission of an alpha-particle (symbol α) to an isotope of uranium (symbol U). The half-life for this decay is 2.4×10^4 years.

- (i) Complete the equation for this decay.



- (ii) The count rate measured due to a pure sample of plutonium-239 is 6400 counts / second.

Calculate the count rate after a period of 1.2×10^5 years has elapsed.

count rate = [3]

- (iii) Fig. 10.1 shows an alpha-particle entering a uniform magnetic field.



Fig. 10.1

1. On Fig. 10.1, draw the path of the alpha-particle in the magnetic field. [2]

2. Explain why the alpha-particle follows the path you drew in (b)(iii)1.

.....
 [1]

(c) In a nuclear reactor at a power station, plutonium-239 is made to undergo nuclear fission.

(i) State what is meant by *nuclear fission*.

.....
..... [1]

(ii) State what happens to a plutonium-239 nucleus in the nuclear reactor to cause fission.

.....
..... [1]

(iii) State one advantage of generating electricity in a nuclear power station rather than using

1. an oil-fired power station,

.....
..... [1]

2. an array of wind turbines.

.....
..... [1]

22. M/J/18/22/Q11

A smoke detector contains a small radioactive source of americium-241. The source emits alpha-particles.

- (a) (i) In the space below, draw a diagram of an apparatus that can be used to show that the source emits alpha-particles.

[2]

- (ii) Describe how this apparatus is used.

.....

.....

.....

.....[2]

- (iii) Explain how the results of the test show that the source emits alpha-particles.

.....

.....

.....

.....[2]

- (b) The smoke detector works because alpha-particles from the source ionise the air.

Compare the relative ionising effects and penetrating powers of alpha-particles, beta-particles and gamma rays.

ionising effects

.....

penetration effects

.....[2]

(c) Americium-241 has a half-life of 430 years.

(i) Radium-224 is another element that emits alpha-particles. It has a half-life of 3.6 days.

Explain why americium-241 is more suitable as the radioactive source in a smoke detector than radium-224.

.....
.....
.....[1]

(ii) A smoke detector contains 8.0×10^{11} atoms of americium-241.

Calculate the time taken for the number of atoms of americium-241 to fall to 1.0×10^{11} .

time =[3]

(d) When used correctly, the radioactive source in the smoke detector is less harmful than background radiation.

(i) State one harmful effect of background radiation.

.....[1]

(ii) A radioactive source is picked up using a long-handled tool. Explain why this is safer than using a short-handled tool.

.....
.....
.....
.....[2]

23. O/N/17/21/Q8

A sample of the radioactive isotope radon-222 decays by the emission of alpha-particles.

(a) As alpha-particles travel through air, the air is ionised.

(i) Describe the composition of an alpha-particle.

.....
.....
..... [1]

(ii) Explain how an alpha-particle ionises air.

.....
.....
..... [2]

(iii) State how the relative ionising effect of alpha-particles compares with that of

1. beta-particles,

.....

2. gamma rays.

..... [1]

(b) The half-life of radon-222 is 3.3×10^5 s.

The number of radon-222 atoms in the sample is 4.8×10^4 .

(i) Determine the time that it takes for the number of radon-222 atoms in this sample to decrease to 1.5×10^3 .

time = [3]

(ii) Suggest one reason why, in practice, the time for the number of radon-222 atoms to decrease to 1.5×10^3 may differ slightly from the value obtained in **(b)(i)**.

.....
..... [1]

24. M/J/17/22/Q11

One source of background radiation is cosmic rays.

The cosmic rays that enter the Earth's atmosphere are known as primary cosmic rays. Most of these rays are fast-moving protons or fast-moving alpha-particles.

Both the Earth's magnetic field and the Earth's atmosphere reduce the number of primary cosmic rays that reach the Earth's surface.

- (a) (i)** Name a source of background radiation apart from cosmic rays.

.....
.....[1]

- (ii)** State one effect of background radiation.

.....
.....[1]

- (iii)** State the number of protons and the number of neutrons in an alpha-particle.

number of protons

number of neutrons

[2]

- (iv)** Explain how the Earth's atmosphere reduces the number of alpha-particles that reach the surface.

.....
.....[1]

- (v)** Fig. 11.1 shows an area where there is a uniform magnetic field into the page.

On Fig. 11.1, draw a diagram to show the deflection of an alpha-particle by this magnetic field.

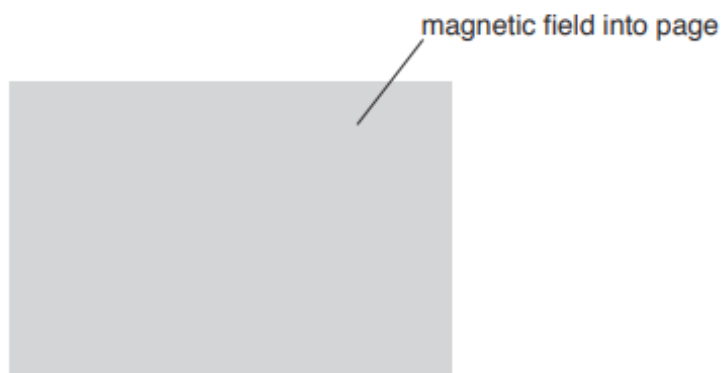


Fig. 11.1

[2]

- (b) Primary cosmic rays produce carbon-14 in the atmosphere. The half-life of carbon-14 is 5700 years.

(i) Define the term *half-life*.

.....
.....[2]

- (ii) A specimen of wood is found to contain 25 000 atoms of carbon-14. An otherwise identical specimen taken from a living tree is found to contain 200 000 atoms of carbon-14.

Determine the age of the specimen that contains 25 000 atoms of carbon-14.

age =[2]

- (iii) State why it is not possible to use the amount of carbon-14 to date a specimen that is 570 000 years old.

.....
.....[1]

- (c) Carbon-14 is one of the isotopes of carbon. Another isotope is carbon-12.

(i) Describe how a nucleus of carbon-14 differs from a nucleus of carbon-12.

.....
.....
.....[2]

(ii) Describe one similarity in the structure of the nuclei of these isotopes.

.....
.....
.....[1]

25. M/J/17/21/Q8

A doctor has the choice of two isotopes, $^{131}_{53}\text{I}$ and $^{125}_{53}\text{I}$, to inject into a patient.

These isotopes emit radiation from deep inside the body. The radiation is detected outside the body and provides information to help the doctor find out what is wrong with the patient.

The isotope $^{131}_{53}\text{I}$ decays into a stable isotope of xenon by emitting a beta-particle (β) and a gamma ray (γ).

(a) State what is meant by a *gamma ray*.

.....
.....[1]

(b) Complete the equation to show the radioactive decay of $^{131}_{53}\text{I}$.



[2]

(c) When a nucleus of $^{125}_{53}\text{I}$ decays, only a gamma ray is emitted.

Suggest why it is better for the doctor to inject the isotope $^{125}_{53}\text{I}$ into the patient.

.....
.....
.....
.....[2]

26. O/N/16/22/Q7

Plutonium-238 is a radioactive isotope that decays by alpha-particle emission. It also emits gamma-rays.

- (a) The nuclide notation for plutonium-238 is ${}_{94}^{238}\text{Pu}$.

Describe the structure and composition of a neutral atom of plutonium-238.

.....

.....

.....

.....[3]

- (b) Alpha-particles, beta-particles and gamma-rays all cause ionisation. The ionisation of substances in the human body can be hazardous.

(i) State how the ionising effect of beta-particles compares with

1. the ionising effect of alpha-particles,

.....[1]

2. the ionising effect of gamma-rays.

.....[1]

(ii) State two precautions that are taken when radioactive substances are moved in a safe way.

1.

.....

2.

.....

[2]

27. O/N/16/21/Q8

Sodium-24 and sodium-23 are two of the isotopes of sodium. Sodium-24 is a radioactive isotope that emits beta-particles and gamma-rays as it decays.

(a) The nuclide notation for the isotope sodium-24 is ${}^{24}_{11}\text{Na}$.

(i) Describe the composition and structure of a neutral atom of sodium-24.

.....

.....

.....

.....

.....[3]

(ii) State how the composition of a nucleus of sodium-24 differs from the composition of a nucleus of sodium-23.

.....

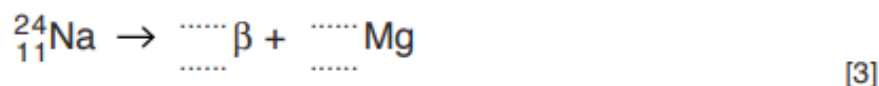
.....[1]

(b) A nucleus of sodium-24 decays. It emits a beta-particle and produces a nucleus of an isotope of magnesium (Mg).

(i) State the name of the particle that is identical to a beta-particle.

.....[1]

(ii) Complete the nuclide equation for the emission of a beta-particle (β) by sodium-24.



(c) When it decays, sodium-24 also emits gamma-rays.

State what gamma-rays are.

.....

.....

.....[2]

- (d) Fig. 8.1 shows a beam of beta-particles and a beam of gamma-rays entering the electric field between two oppositely charged plates.

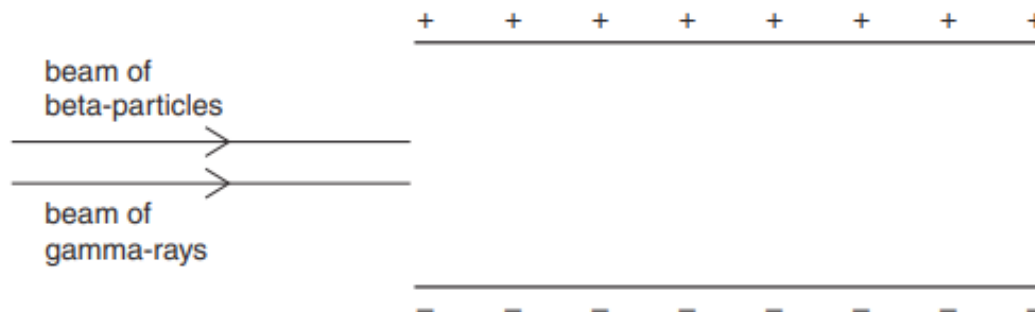


Fig. 8.1 (not to scale)

- (i) On Fig. 8.1, draw the path of the beta-particles in the electric field between the charged plates. [1]
- (ii) On Fig. 8.1, draw the path of the gamma-rays in the electric field between the charged plates. [1]
- (iii) Explain why the path taken by the beta-particles differs from that taken by the gamma-rays.
-[1]
- (e) Sodium-24 is sometimes used in diagnosing medical conditions. The half-life of sodium-24 is 15 hours.

Explain one reason why a half-life of 15 hours makes the isotope suitable for use in the human body.

.....[2]

28. M/J/16/22/Q11

Fig. 11.1 represents a nuclear fusion reaction.

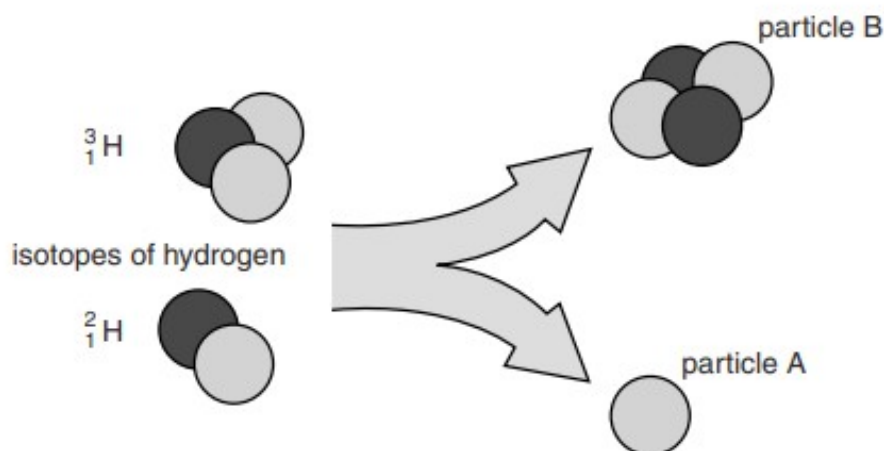


Fig. 11.1

At very high temperatures, nuclei of the two isotopes ${}^2_1\text{H}$ and ${}^3_1\text{H}$ fuse together. Energy is produced and two new particles are formed, particle A and particle B.

(a) Explain what is meant by *isotopes* of hydrogen.

.....

[2]

(b) Using Fig. 11.1, state

- (i) the number of neutrons in a nucleus of the isotope ${}^3_1\text{H}$,
- (ii) the name of particle A,
- (iii) the proton number of particle B,
- (iv) the nucleon number (mass number) of particle B.

[4]

(c) A very high temperature is needed to force nuclei together.

Explain why.

.....

[2]

- (d) A star forms from a large cloud of gas and dust in space.

Describe what happens as the star forms.

.....

.....

.....

.....

.....[3]

- (e) The isotope of hydrogen ${}^3_1\text{H}$ has a half-life of 12 years.

- (i) State what is meant by *half-life*.

.....

.....[2]

- (ii) A sample contains 16 000 atoms of ${}^3_1\text{H}$.

Calculate the number of atoms of ${}^3_1\text{H}$ present in the sample after 48 years.

number =[2]

29. O/N/15/22/Q11

The proton number (atomic number) of the element lead is 82. The isotope lead-209 ($^{209}_{82}\text{Pb}$) is radioactive and decays by the emission of beta-particles.

(a) Describe the composition and structure of a neutral atom of lead-209.

.....

.....

.....

..... [3]

(b) A nucleus of lead-209 emits a beta-particle.

(i) State how the composition of the nucleus produced differs from the original nucleus.

.....

..... [2]

(ii) Complete the table in Fig. 11.1 to show the relative ionising effects of the three types of ionising radiation produced by radioactive decay.

least strongly ionising $\longrightarrow$ most strongly ionising		

Fig. 11.1

[2]

(iii) A sample containing lead-209 nuclei is used to produce a beam of beta-particles. The beta-particles enter a magnetic field. The magnetic field is perpendicular to the direction of travel of the beta-particles.

The beta-particles travel from left to right. Fig. 11.2 shows that the direction of the magnetic field is out of the page.



Fig. 11.2

On Fig. 11.2, sketch the path of the beta-particles in the magnetic field.

[2]

(c) An industrial technician uses a detector to measure the background count rate in a laboratory.

(i) State what is meant by *background radiation*.

.....
..... [1]

(ii) Suggest two major sources of background radiation.

1.

2.

[2]

(iii) The average reading for the background count measured by the technician is 16 counts/minute.

He then brings a sample that contains a radioactive isotope of lead close to the detector and he finds that the new count rate is 92 counts/minute. The half-life of this isotope is 3.3 hours.

Determine the count rate measured using the detector after 6.6 hours.

count rate = [3]

30. O/N/15/21/Q11

Fig. 11.1 shows the proton number and the nucleon number of the nuclei of five neutral atoms.

	neutral atom 1	neutral atom 2	neutral atom 3	neutral atom 4	neutral atom 5
proton number	16	17	18	18	19
nucleon number	35	35	38	40	39

Fig. 11.1

(a) State the two atoms in Fig. 11.1 that

(i) are different isotopes of the same element,

..... [1]

(ii) contain the same number of neutrons,

..... [1]

(iii) contain the same number of electrons.

..... [1]

(b) The nucleus of atom 1 is radioactive and when it decays, it emits a beta-particle.

State, for the nucleus produced by this decay

(i) the proton number,

proton number = [1]

(ii) the nucleon number.

nucleon number = [1]

- (c) A sample contains a large number of radioactive nuclei that emit beta-particles. The sample is placed near to a radiation detector in a laboratory and the count rate is recorded over a long period of time.

Fig. 11.2 shows the count rate recorded by the detector during this period.

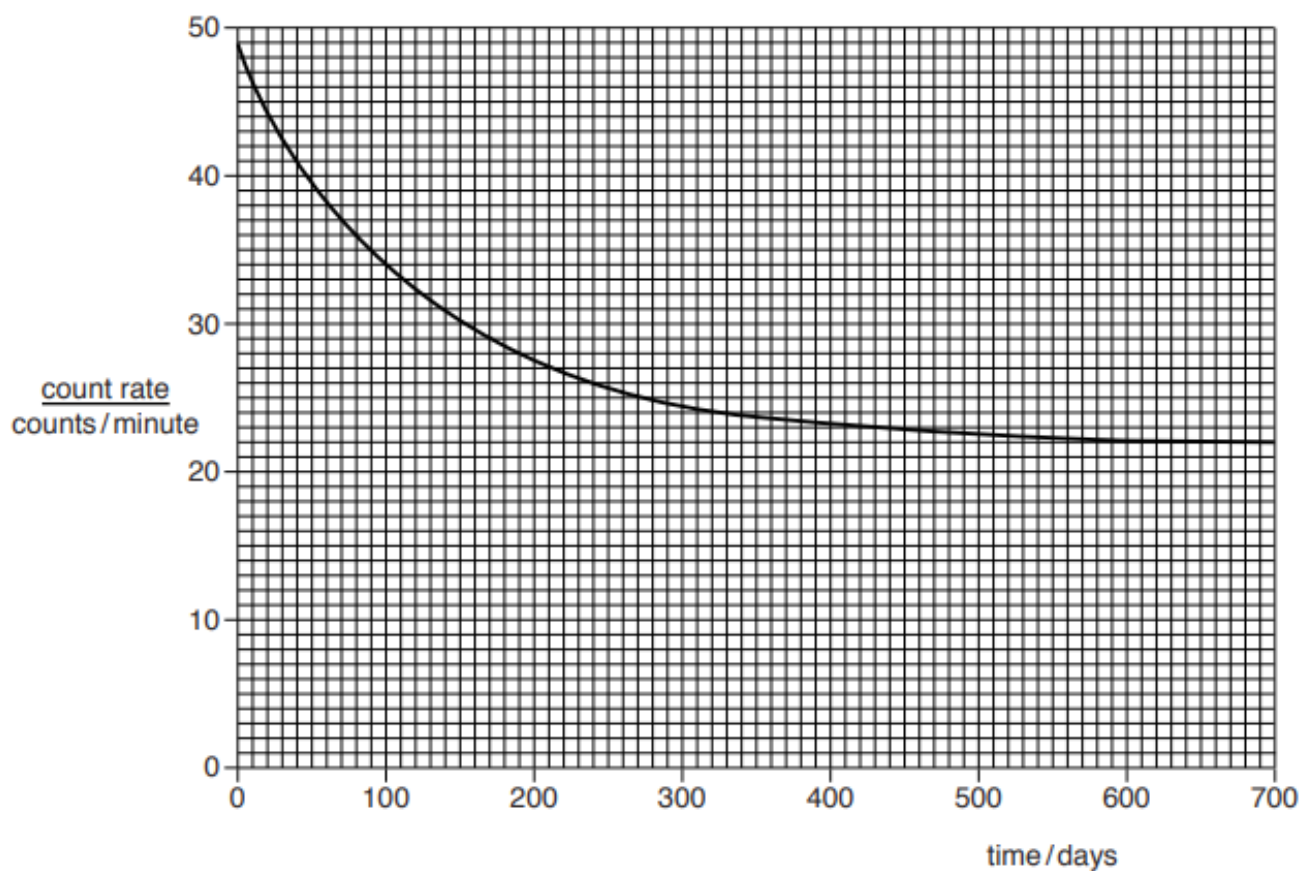


Fig. 11.2

- (i) Suggest two major sources of background radiation.

1.

2.

[2]

(ii) Using Fig. 11.2,

1. determine a value for the background count rate in the laboratory,

background count rate = [1]

2. determine the count rate, at time = 0, due to the sample on its own,

count rate due to sample = [1]

3. estimate the half-life of the radioactive atoms in the sample.

half-life = [2]

(d) A student suggests a change to the experiment in (c). He suggests completely encasing the radiation detector in lead before the sample that emits beta-particles is placed near to it. The effect on the count rate of background radiation may then be ignored.

- (i) Describe and explain the effect on the background count rate recorded by the detector of completely encasing the radiation detector in lead.

.....
.....
..... [2]

- (ii) State and explain whether the student's suggestion is a sensible one.

.....
.....
..... [2]

31. M/J/15/21/Q11

- (a) A student makes a model of an atom. The model contains 24 electrons, 25 protons and 26 neutrons. Some of these particles are inside a nucleus at the centre of the model.

- (i) Determine the nucleon number (mass number) of the atom.

.....[1]

- (ii) Explain why the model represents a charged atom.

.....

[2]

- (iii) The student makes a new model of a different isotope of the same element.

Describe the nucleus of this new model.

.....

[2]

- (b) Americium-241 is radioactive. Its nuclide notation is $^{241}_{94}\text{Am}$.

- (i) Determine the number of neutrons in a nucleus of americium-241.

number =[1]

- (ii) A nucleus of americium-241 emits an α -particle and decays to uranium-237.

Complete the nuclear equation for the decay of americium-241.



- (c) Geiger and Marsden studied the structure of gold atoms. Fig. 11.1 shows a version of their apparatus. Alpha-particles strike a thin gold foil.

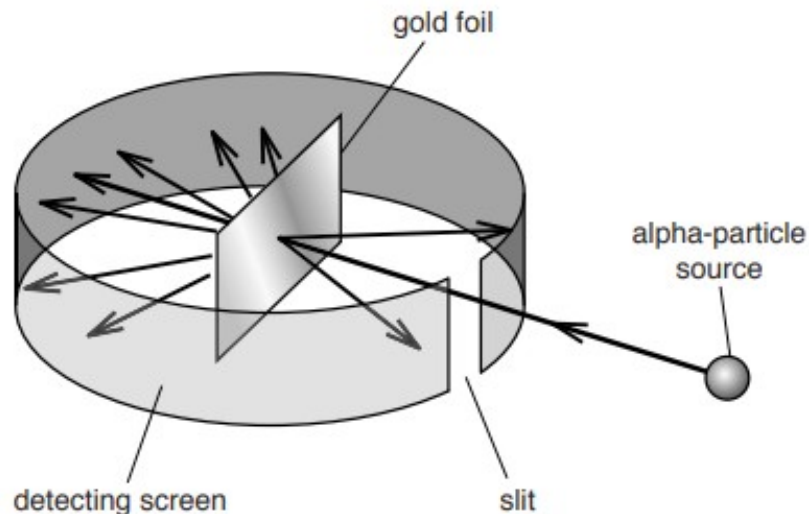


Fig. 11.1

The apparatus shown is in a container from which all the air is removed.

- (i) Suggest why it is necessary to remove all the air from the container.

.....
.....
.....[1]

- (ii) The alpha-particles are emitted from the source at random.

Explain why most of the alpha particles from the source do not reach the gold foil.

.....
.....[1]

- (iii) Fig. 11.2 shows a model of an atom of the gold foil, with its nucleus at the centre.

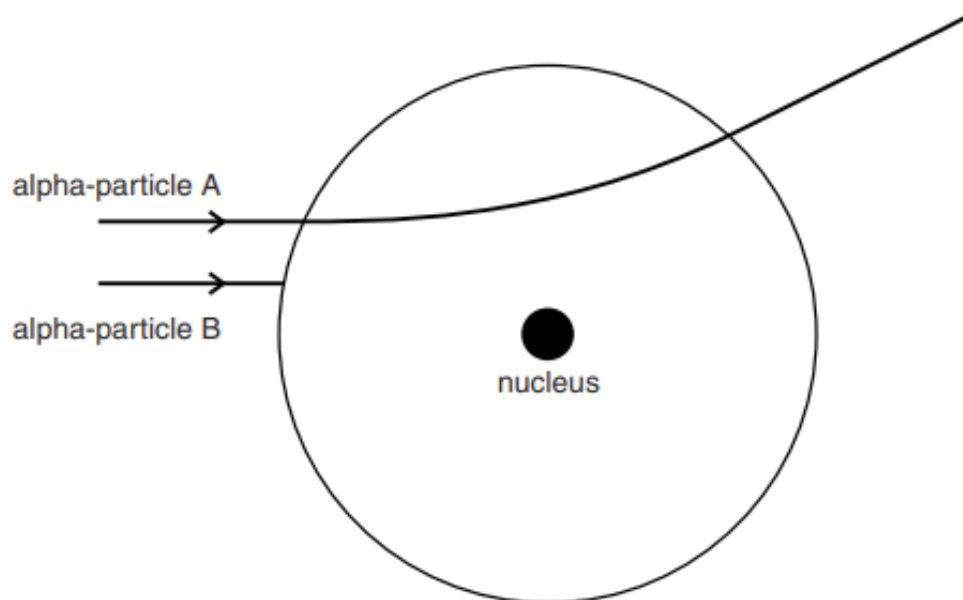


Fig. 11.2 (not to scale)

The alpha-particle labelled A is deflected by the nucleus, as shown.

On Fig. 11.2, complete the path of the alpha-particle labelled B.

[1]

- (iv) Explain how the alpha-particle scattering experiment provides evidence for the existence of a small nucleus inside the atom.

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

32. O/N/14/22/Q11

Strontium-90 ($^{90}_{38}\text{Sr}$) is a radioactive isotope. A nucleus of strontium-90 decays by the emission of a beta-particle (β).

(a) Define the term *isotope*.

.....
[2]

(b) For a neutral atom of strontium-90, state

(i) the number of electrons,

number of electrons =[1]

(ii) the number of neutrons.

number of neutrons =[1]

(c) When a nucleus of strontium-90 decays, it becomes a nucleus of yttrium (symbol Y).

Complete the equation for this decay.



(d) A sample initially contains 6.0×10^8 atoms of strontium-90.

(i) The half-life of strontium-90 is 29 years.

Calculate the number of strontium-90 atoms that remain in the sample after 87 years.

number of atoms =[2]

(ii) Describe an experiment that shows that the sample emits only beta-particles.

.....
.....
.....
.....
.....
.....[4]

(iii) A teacher says that the emission of beta-particles from strontium-90 is random.

1. Explain what is meant by *random* in this context.

.....
.....[1]

2. State **two** ways in which the emission of beta-particles from strontium-90 is random.

.....
.....[2]

33. O/N/14/21/Q8

The nuclide notation for the radioactive isotope boron-12 is ${}^{12}_{5}\text{B}$.

- (a) In the space below, draw a labelled diagram to illustrate the structure of a neutral atom of this isotope. Show all the particles in the atom.

[4]

- (b) As boron-12 decays, it emits a beta-particle. A new atom is produced.

Determine

- (i) the proton number (atomic number) of the new atom,

proton number =[1]

- (ii) the nucleon number (mass number) of the new atom.

nucleon number =[1]

34. M/J/14/22/Q8

- (a) An atom consists of electrons surrounding a nucleus made up of protons and neutrons.

State which of these particles

- (i) have an equal and opposite charge,

..... [1]

- (ii) have almost equal mass.

..... [1]

- (b) The nuclide notation for carbon-14 is $^{14}_6\text{C}$. Carbon-14 decays by beta emission to a stable isotope of nitrogen (N).

- (i) Write numbers in the empty boxes below to show the nuclide notation for this isotope of nitrogen.



[2]

- (ii) The half-life of carbon-14 is 5700 years.

A sample of wood from a living tree contains 2.4×10^{12} atoms of carbon-14.

A similar sample of the same size is taken from an old piece of wood. It contains 6.0×10^{11} atoms of carbon-14.

Calculate the age of the old piece of wood.

age = [2]

35. M/J/14/21/Q8

A hospital laboratory uses a small sample of a radioactive isotope of iodine, $^{131}_{53}\text{I}$.

- (a) (i)** Describe the structure of the nucleus of an atom of this isotope.

.....
 [2]

- (ii)** The sample is radioactive. Describe what happens in radioactive decay.

.....

 [2]

- (b)** The count in one minute from the source is measured several times. Fig. 8.1 shows the readings obtained.

2686	2759	2847	2799
------	------	------	------

Fig. 8.1

- (i)** Suggest why the readings are different.

.....
 [1]

- (ii)** The half-life of $^{131}_{53}\text{I}$ is 8.0 days. Estimate the count in one minute obtained from the sample after 24 days.

count = [2]

36. O/N/13/22/Q8

A smoke detector containing an alpha-particle source is fixed to the ceiling in a room. The alpha-particles ionise the air between two metal plates so that a battery produces a current between the plates.

If smoke enters the detector, it reduces the current between the plates and an alarm sounds.

Fig. 8.1 shows the arrangement.

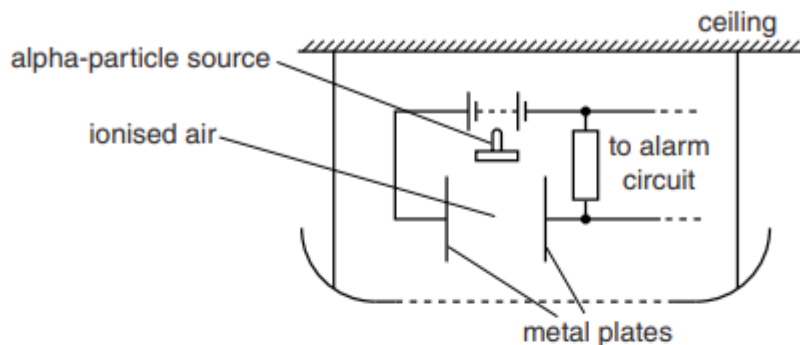


Fig. 8.1

- (a) A technician removes the source from the smoke detector and measures the count rate of the source.

- (i) Suggest two precautions that the technician should take when working with the source.

1.
2.

[2]

- (ii) The technician measures the count rate, in counts/s, five times. On each occasion, the arrangement of the equipment is the same.

The values obtained are: 81 77 80 83 and 79.

Suggest why the values are different.

-
.....[1]

- (b) Radioactive sources that emit beta-particles or gamma-rays are not used in smoke detectors. State and explain two reasons for this.

1.
.....
2.
.....

[3]

37. O/N/13/21/Q8

A radioactive sample emits only beta-particles and gamma-rays.

- (a) Describe, with the aid of a labelled diagram, an experiment that shows that the sample does **not** emit alpha-particles.

.....

.....

.....

.....

..... [3]

- (b) State two safety precautions that must be taken when working with the radioactive sample.

1.

2. [2]

38. M/J/13/21/Q11

(a) The nucleus of an atom of carbon-14 is represented as $^{14}_6\text{C}$.

(i) Describe the structure of an atom of carbon-14.

.....
.....
..... [3]

(ii) Carbon has several isotopes.

Describe, in terms of the particles within the nucleus, how isotopes are

1. different from one another,

..... [1]

2. the same as one another.

..... [1]

(b) Carbon-14 decays by beta emission to a stable isotope of nitrogen.

Carbon-14 has a half-life of 5700 years and is used to find the age of objects by carbon dating.

(i) A pure sample of carbon-14 initially contains 8.0×10^{20} atoms.

Calculate the number of atoms of carbon and of nitrogen present in the sample after 11 400 years.

number of atoms of carbon =

number of atoms of nitrogen = [3]

(ii) Explain why carbon-14 is not used to find the age of a piece of coal that has been in the ground for millions of years.

.....
..... [1]

- (c) A teacher sets up the apparatus shown in Fig. 11.1.

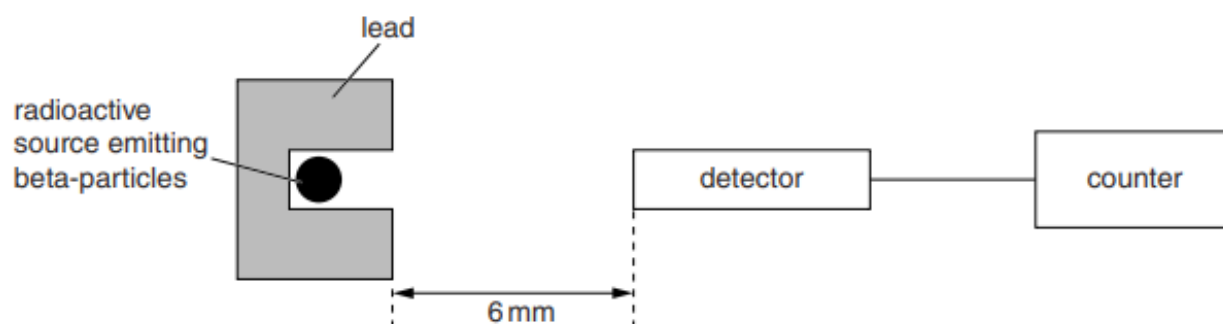


Fig. 11.1

The teacher also has available a number of sheets of aluminium, each of which is 1 mm thick.

- (i) Describe how the apparatus and the sheets are used to demonstrate that beta-particles are stopped by 5 mm of aluminium.

In your account, make clear how the background count is measured and used.

.....

.....

.....

.....

.....

..... [4]

- (ii) Suggest why the lead in Fig. 11.1 is placed around the radioactive source during the demonstration.

.....

..... [1]

- (iii) The radioactive source in Fig. 11.1 is changed for one that emits gamma-rays. Describe why the apparatus cannot be used to find the thickness of aluminium needed to stop gamma-rays.

.....

..... [1]

39. O/N/12/22/Q7

A small piece of metal taken from a nuclear reactor is radioactive.

(a) Describe a method to determine whether the piece of metal emits gamma-rays.

.....

.....

.....

.....

.....

..... [3]

(b) State two safety precautions that must be taken when handling the piece of metal.

1.

.....

2.

.....

[2]

40. M/J/12/22/Q10

To find out whether a radioactive source emits alpha-particles, beta-particles or gamma-rays, the source is placed in front of a radiation detector, as shown in Fig. 10.1.

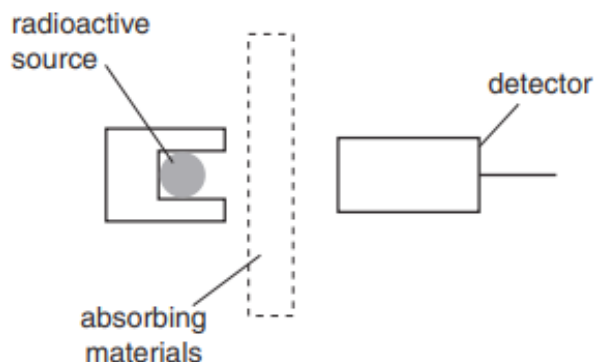


Fig. 10.1

Different absorbing materials are placed between the source and the detector. The detector measures the number of counts per minute.

(a) State what is meant by

(i) an alpha-particle,

.....
.....[2]

(ii) a gamma-ray.

.....
.....[2]

(b) (i) Suggest why, in this experiment,

1. the distance between the source and the detector is only a few centimetres,

.....
.....[2]

2. the half-life of the radioactive source is longer than a few minutes.

.....
.....[1]

(ii) State one precaution taken when using a radioactive source.

.....
.....[1]

(c) Fig. 10.2 shows the results obtained.

source present	material between source and detector	counts per minute
no	none	10
yes	none	1200
yes	thin paper	820
yes	5 mm aluminium	820

Fig. 10.2

Using information from Fig. 10.2, state and explain whether the source emits

(i) alpha-particles,

.....
[1]

(ii) beta-particles,

.....
[2]

(iii) gamma-rays.

.....
[1]

(d) There is a count recorded even when no source is present. This is caused by background radiation.

State two sources of background radiation.

1.
 2.
 [2]

(e) Describe one effect on the human body of a very high level of radiation.

.....
[1]

41. M/J/12/21/Q11

Fig. 11.1 shows part of a smoke detector.

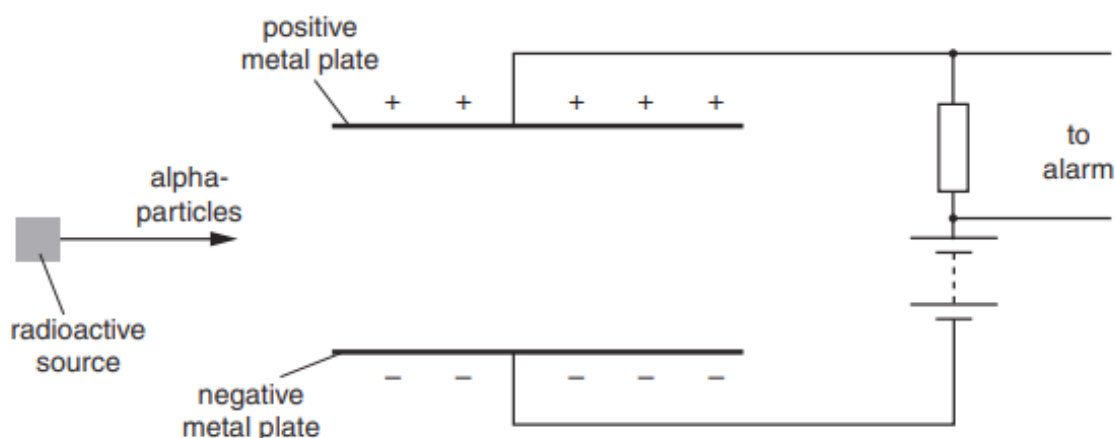


Fig. 11.1 (not to scale)

Alpha-particles from the radioactive source ionise the air between the metal plates. There is then a small current in the resistor. When smoke enters the detector, the current changes and the alarm sounds.

- (a) Explain why a source emitting only gamma-rays is unsuitable for use in the smoke detector.

.....
[1]

- (b) There is an electric field between the metal plates.

- (i) On Fig. 11.1, draw the electric field between the plates. [3]

- (ii) The alpha-particles in Fig. 11.1 are deflected slightly as they pass through the electric field.

Explain why the alpha-particles are deflected and state the direction of the deflection.

.....

[2]

- (c) State and explain how the path of gamma-rays differs from the path of alpha-particles as they pass through an electric field.

.....

[2]

- (d) The source used is the radioactive isotope americium-241, which has a half-life of 500 years.
- (i) A sample of americium-241 is left for 1500 years. Determine the fraction of the original number of atoms of americium-241 left in the sample after this time.

fraction =[2]

- (ii) Another isotope of the same element, americium-242, has a half-life of 16 hours. Explain why the half-life of americium-242 makes this isotope unsuitable for use in a smoke detector.

.....
.....
.....[1]

- (iii) Atoms of americium-241 and americium-242 contain protons, neutrons and electrons.

1. State **two** similarities in the structure of an atom of americium-241 and an atom of americium-242.

.....
.....
.....[2]

2. Describe the difference between an atom of americium-241 and an atom of americium-242.

.....
.....
.....[2]

42. O/N/11/22/Q11

Fig. 11.1 illustrates a neutron hitting a uranium-235 nucleus.

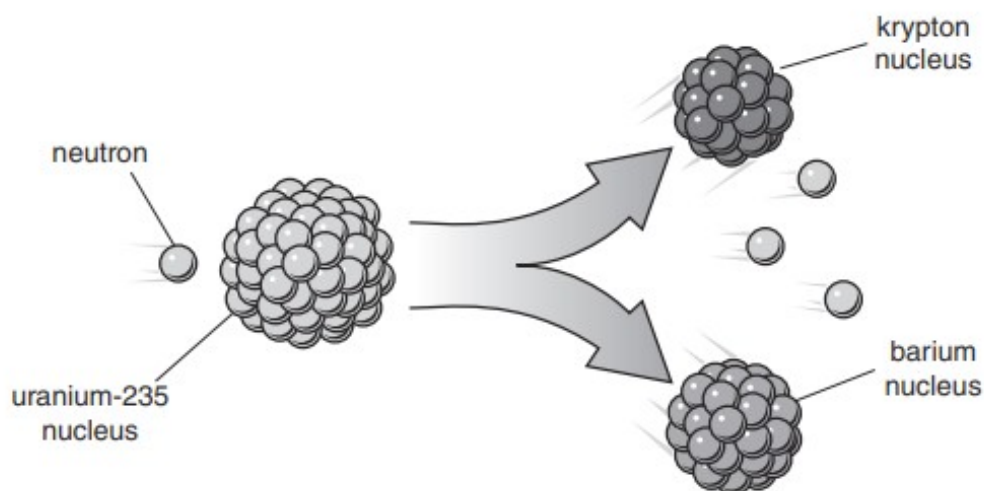


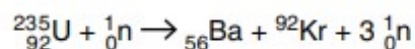
Fig. 11.1

The uranium-235 nucleus splits into a nucleus of barium (Ba) and a nucleus of krypton (Kr) and three neutrons are released.

(a) State the name of this process.

.....[1]

(b) This process may be represented by a nuclear equation. An incomplete version of this equation is shown below.



(i) Calculate

1. the number of neutrons in a nucleus of uranium-235,

number of neutrons =[1]

2. the proton number (atomic number) of krypton,

proton number =[1]

3. the nucleon number (mass number) of the barium nucleus.

nucleon number =[1]

- (ii) During this process, there is a decrease in mass. Energy is released. The decrease in mass is 3.1×10^{-28} kg and the speed of light is 3.0×10^8 m/s.

Calculate the energy released in this process.

energy =[3]

- (c) A nuclear power station generates electrical energy. In the power station, steam is used to drive a turbine. Describe how the splitting of uranium-235 is used to produce the steam. You may include a block diagram.

.....
.....
.....
.....
.....[5]

- (d) Some of the waste products from a nuclear power station are radioactive and have very long half-lives.

- (i) State the meaning of *half-life*.

.....
.....[2]

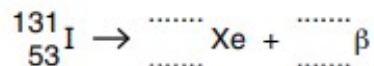
- (ii) Describe one safety precaution that is taken when radioactive waste products are handled.

.....
.....[1]

43. O/N/11/21/Q8

- (a) One radioactive isotope of iodine is $^{131}_{53}\text{I}$. As a nucleus of this isotope decays, it emits a beta-particle (symbol: β) and it becomes a nucleus of an isotope of xenon (symbol: Xe).

Complete the equation in nuclide form, for this decay.



[2]

- (b) A sample of a radioactive isotope emits both beta-particles and gamma-rays.

Fig. 8.1 shows these two types of radiation entering a magnetic field.

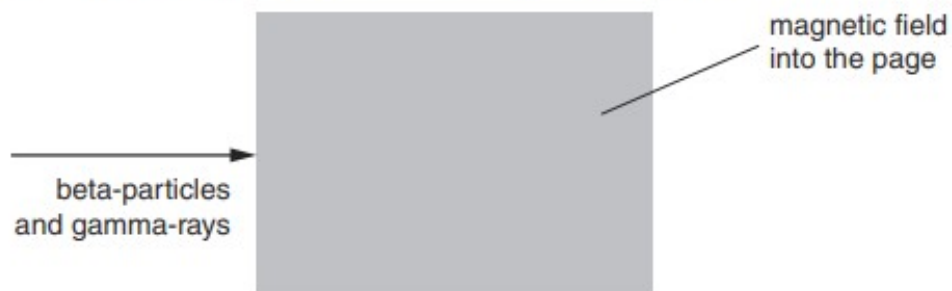


Fig. 8.1

The direction of the magnetic field is into the page.

On Fig. 8.1, mark **and** label the path taken by

- (i) the beta-particles, [1]
- (ii) the gamma-rays. [1]
- (c) Emission from a radioactive source is a random process.

State two ways in which the process is random.

1.
2.

[2]

44. M/J/11/22/Q8

Fig. 8.1 illustrates the process that occurs in the core of a nuclear reactor.

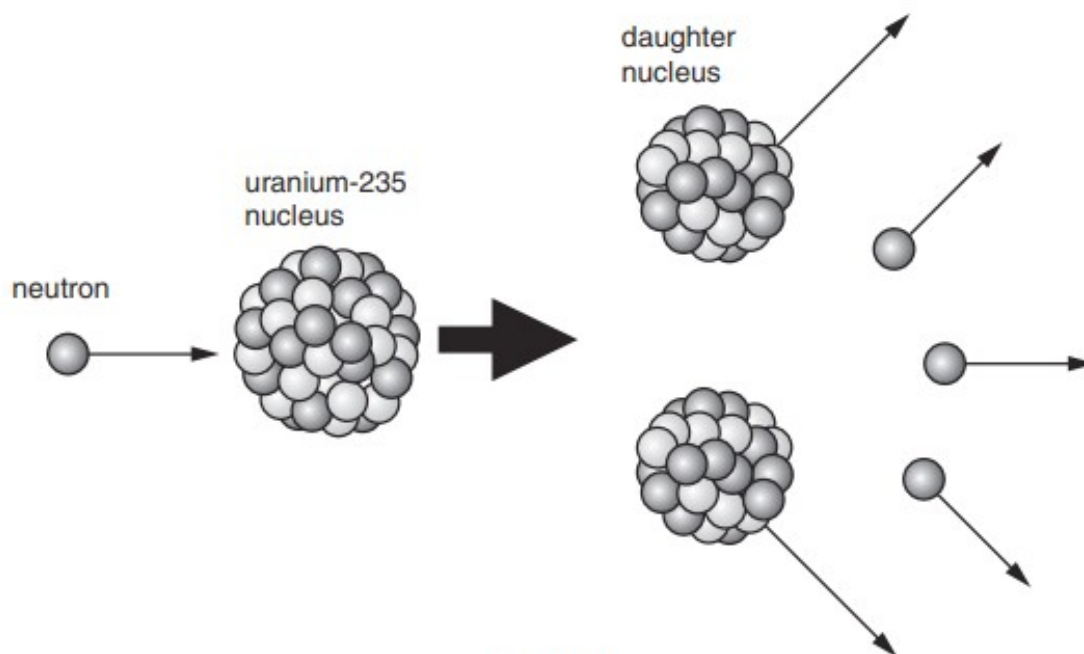


Fig. 8.1

(a) State the name of the process illustrated in Fig. 8.1.

.....[1]

(b) Describe what happens during this process.

.....
.....
.....[2]

(c) Some of the waste from a nuclear reactor is radioactive with a long half-life.

Explain what is meant by

(i) *radioactive*,

.....
.....[1]

(ii) *a long half-life*.

.....
.....[2]

45. M/J/11/21/Q11

An atom of one isotope of sodium contains 11 protons, 13 neutrons and 11 electrons.

- (a) (i) State the nucleon number (mass number) of this isotope.

..... [1]

- (ii) Explain why the number of protons is equal to the number of electrons in a neutral atom.

.....

..... [1]

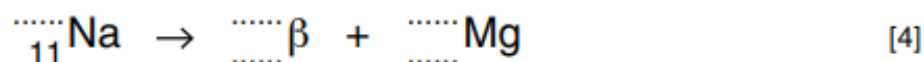
- (b) A nucleus of this isotope undergoes radioactive decay by the emission of a beta-particle.

- (i) State the energy change that occurs during radioactive decay.

.....

..... [2]

- (ii) Complete the nuclear equation that represents this decay by filling in the appropriate numbers in the blank spaces. Na is the chemical symbol for sodium.



- (c) At the start of an experiment, the count rate due to a sample of this isotope of sodium is found to be 1200 counts per minute. This is shown on Fig. 11.1 by a point plotted at time $t = 0$.

The half-life of this isotope is 15 hours.

- (i) On Fig. 11.1, draw a smooth line to show how the count rate is expected to change between $t = 0$ and $t = 30$ hours. [3]

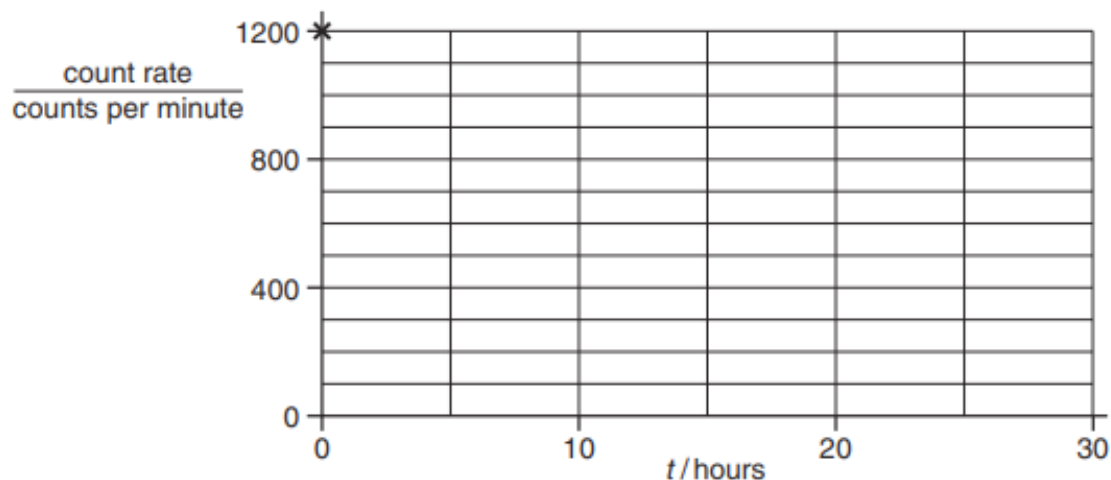


Fig. 11.1

- (ii) Explain why readings taken in an actual experiment may not follow a smooth line.

.....
..... [1]

- (iii) Determine the count rate due to this isotope 30 hours before the start of the experiment.

count rate = [1]

- (iv) In the space below, draw a labelled diagram of the apparatus used to measure the count rate of the radioactive isotope.

[2]

46. O/N/10/21/Q10

An atom of phosphorus (chemical symbol P) is composed of 15 electrons, 17 neutrons and 15 protons.

(a) For this atom, state

(i) the proton number (atomic number), [1]

(ii) the nucleon number (mass number). [1]

(b) This phosphorus atom is radioactive. As it decays to an atom of sulfur (chemical symbol S), it emits a beta-particle (symbol β).

(i) Write a nuclear equation for this decay.

[3]

(ii) Explain what is meant by a *beta-particle*.

.....

..... [2]

(iii) A sample of radioactive material contains many of these phosphorus atoms. Describe an experiment to investigate whether the sample emits only beta-particles. A diagram may be included.

.....

.....

.....

.....

.....

.....

..... [4]

(c) This isotope of phosphorus has a radioactive half-life of 14.3 days.

(i) Explain what is meant by *radioactive half-life*.

.....
.....
..... [2]

(ii) A solution containing such phosphorus atoms is used in a medical procedure. When it is prepared, the solution has an activity of 1400 counts/second. The solution cannot be used when its activity falls below 350 counts/second.

Calculate the maximum time between preparing and using the solution.

time = [2]

47. M/J/10/22/Q8

A radioactive rock emits gamma-rays. A teacher plans an experiment to show a class that the emission of gamma-rays from the rock is random in time.

(a) (i) State the apparatus that the teacher needs to detect the gamma-rays.

.....
.....[1]

(ii) State one safety precaution that the teacher must take.

.....
.....[1]

(iii) Describe how the experiment is performed.

.....
.....
.....
.....
.....
.....[2]

(b) Describe what is meant by a *gamma-ray*.

.....
.....[2]

48. M/J/10/21/Q11

(a) Complete the table of Fig. 11.1.

type of radiation	nature of radiation	charge
alpha-particle	two protons and two neutrons	positive
beta-particle		
gamma-ray		

Fig. 11.1

[4]

(b) Technetium-99 is a radioactive isotope that emits gamma-rays and it has a half-life of 6.0 hours.

In a hospital, a solution containing this isotope is injected into a patient. The solution travels around the body and the isotope attaches to cancer cells. A few hours after the injection, an instrument is used to detect the gamma-rays emitted by the isotope. A high reading on the instrument indicates the position of cancer cells.

(i) Explain what is meant by *half-life*.

.....

[2]

(ii) Explain why an isotope emitting gamma-rays is used rather than an isotope emitting alpha-particles.

.....

[2]

49. O/N/09/Q8

(a) State the number of

(i) electrons in a neutral atom of the isotope ${}^7_3\text{Li}$,

number of electrons = [1]

(ii) nucleons in an atom of the isotope ${}^{208}_{82}\text{Pb}$,

number of nucleons = [1]

(iii) protons in an atom of the isotope ${}^{23}_{11}\text{Na}$.

number of protons = [1]

(b) An atom of the isotope chlorine-35 contains 17 protons.

For an atom of the isotope chlorine-37, state the number of

(i) protons,

number of protons = [1]

(ii) neutrons.

number of neutrons = [1]

50. M/J/09/Q8

The symbol ${}_{92}^{238}\text{U}$ represents a nucleus of uranium with nucleon number (mass number) 238 and proton number (atomic number) 92.

(a) State the meaning of *nucleon number*.

.....
.....
..... [2]

(b) A nucleus of uranium-238 decays to form a nucleus of thorium by the emission of an alpha-particle.

State

(i) the proton number of an alpha-particle,

..... [1]

(ii) the nucleon number of an alpha-particle,

..... [1]

(iii) the proton number of thorium,

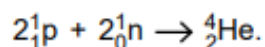
.....

(iv) the nucleon number of the thorium isotope formed.

..... [1]

51. O/N/08/Q8

The reaction that takes place at the centre of the Sun can be represented as



- (a) State the name of this type of reaction.

..... [1]

- (b) The speed of light is $3.0 \times 10^8 \text{ m/s}$.

- (i) The mass of the helium nucleus (${}^4_2\text{He}$) produced is different from the total mass of the two protons and the two neutrons. The difference is $6.6 \times 10^{-29} \text{ kg}$.

Explain why energy is released during the reaction.

.....
.....
..... [2]

- (ii) Calculate the energy released in this reaction. State the formula used.

energy = [3]

52. M/J/07/Q8

(a) Explain how it is possible for an element to have different isotopes.

.....

.....

.....

..... [1]

(b) State what is meant by the *half-life* of a radioactive isotope.

.....

.....

.....

..... [2]

(c) Fig. 8.1 shows how the number of atoms of a radioactive isotope changes with time.

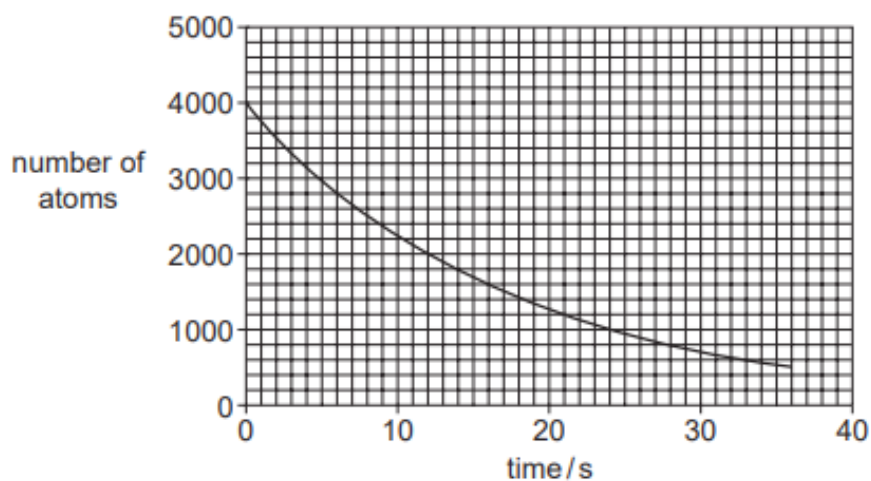


Fig. 8.1

Determine the half-life of the radioactive isotope. On Fig. 8.1, show how you obtained your result.

half-life = [2]

53. M/J/06/Q8

A radioactive source undergoes radioactive decay.

- (a) Explain what is meant by *radioactive decay*.

.....

.....

..... [2]

- (b) Gamma-rays from a radioactive source are used to treat some foods.

The gamma-rays kill bacteria in the food so that it stays fresh.

Some people are worried that food treated in this way becomes radioactive. A scientist tests three food samples. She measures the count-rate of the food before and after treatment with gamma-rays. The results are shown in the table below. The radioactive source is not present during any test.

food sample	count-rate before treatment/ counts per minute	count-rate after treatment/ counts per minute
A	29	32
B	32	28
C	30	31

- (i) Explain why there is a measured count-rate before the food is treated.

.....

.....

.....

..... [1]

- (ii) Determine and explain whether the treated food becomes radioactive.

.....

.....

.....

.....

.....

..... [2]